



# Government procurement and resource misallocation: Evidence from China

Jing Hang<sup>a</sup>, Chaoqun Zhan<sup>b,\*,1</sup>

<sup>a</sup> Lingnan (University) College, Sun Yat-sen University, China

<sup>b</sup> Department of Accountancy, Economics and Finance, Hong Kong Baptist University, Hong Kong Special Administrative Region of China

## ARTICLE INFO

### JEL classification:

H57  
O43  
O47  
O53

### Keywords:

Government procurement  
Resource misallocation  
Corruption

## ABSTRACT

Government often acts as the largest buyer in an economy through government procurement. The allocation of procurement orders however is often distorted. We build a model with heterogeneous firms endogenously seeking government favors in procurement to show that misallocation of procurement orders further leads to misallocation of production factors. Using the enactment of the <Chinese Government Procurement Law> in 2003 as a natural experiment, we find that tighter regulation of procurement in China lowers the extent of resource misallocation in industries that rely more on government procurement. We further provide firm-level evidence in support of our model.

## 1. Introduction

Economists have long been pondering over the large international productivity differences. Recent literature pioneered by Restuccia and Rogerson (2008) and Hsieh and Klenow (2009) points to the importance of resource misallocation between micro production units. Such misallocation not only results from market frictions but also from government actions. Various government policies and regulations have been singled out as sources of resource misallocation.<sup>2</sup> However, policies and regulations are not the only ways through which governments can interfere with resource allocation. In modern economies, one of the important roles that the government plays is as a direct market participant.

The government often acts as the largest buyer of goods and services in an economy. In 2017, OECD countries on average spent 12.8% of GDP on public procurement, for example, Germany spent 15.5%, and the U.S., 9.3%.<sup>3</sup> In China, the country of our study, the share is a bit lower, but it still reached about 4% in 2018.<sup>4</sup> The huge size of government procurement, together with the fact that government agencies often can not effectively monitor the use of public funds due to the lack of incentive and capability, makes it one of the major breeding grounds for corruption (Bosio et al., 2022). A growing literature has shown how corruption in public

\* Corresponding author.

E-mail address: [cqzhan@hkbu.edu.hk](mailto:cqzhan@hkbu.edu.hk) (C. Zhan).

<sup>1</sup> Authors are listed in alphabetical order and contribute equally.

<sup>2</sup> These policy distortions include labor market regulations (Gourio and Roys, 2014; Garicano et al., 2016), product market regulations (Garcia-Santana and Pijoan-Mas, 2014), environmental standards (Tombe and Winter, 2015), and land policies (Adamopoulos and Restuccia, 2020). Andrews et al. (2014) examine the effect of various public policies on resource allocation using firm data from OECD countries. Even if the policies are not distortionary by design, misallocation could result from inefficient enforcement, see Bachas et al. (2019) for evidence of size-dependent tax enforcement.

<sup>3</sup> <https://stats.oecd.org/Index.aspx?QueryId=94406>.

<sup>4</sup> [http://www.cccp.gov.cn/jdjc/fxyj/201909/t20190904\\_12826216.htm](http://www.cccp.gov.cn/jdjc/fxyj/201909/t20190904_12826216.htm).

<https://doi.org/10.1016/j.jebo.2023.10.014>

Received 22 October 2022; Received in revised form 27 September 2023; Accepted 10 October 2023

Available online 7 November 2023

0167-2681/© 2023 Elsevier B.V. All rights reserved.

procurement has led to misallocation of procurement contracts, and therefore the wasting of public funds.<sup>5</sup> Even without corruption, procurement contracts can be misallocated by government agencies to serve purposes other than efficient use of public funds, such as trade protection.<sup>6</sup> This paper moves one step further to ask whether the misallocation of procurement orders distorts the allocation of *production factors* across firms, and quantifies the productivity loss due to such misallocation.

We utilize the enactment of a Chinese law regulating public procurement to test the causal relationship between the misallocation of procurement orders and the misallocation of production factors. The *Government Procurement Law of the People's Republic of China* (the GPL, hereafter), discussed in detail in Section 2, was enacted in January 2003 mainly for improving efficiency in using government procurement funds by standardizing and increasing the transparency of the procurement process.<sup>7</sup> Successful implementation of the GPL should reduce the importance of non-market factors in the allocation of procurement order, and further reduce the misallocation of production factors, which can be traced in data. We empirically examine this link from the GPL to resource misallocation in Chinese manufacturing firms. The results also provide an evaluation of the effectiveness of the GPL, which has important policy implications.

Our empirical work is guided by a theoretical model with heterogeneous firms. In the model, firms sell both in the market and to the government. Firms spend a marketing cost to win government orders. The cost is lower for firms with better connections with the government. Production requires a fixed cost such that firms with higher physical productivity have larger size and higher revenue productivity under efficient allocation.<sup>8</sup> Misallocation of procurement orders lowers the positive covariance between firm size and productivity: better-connected firms raise their size by selling more to the government but lower their productivity due to the marketing cost. Regulation of the procurement process weakens the effects of connections, increasing the size–productivity covariance. The model thus naturally suggests the use of the covariance as a measure of resource misallocation empirically, which is first proposed by Bartelsman et al. (2013). Industries differ in their reliance on government procurement. Our model predicts that strengthened regulation of government procurement due to the GPL should improve resource allocation in industries that sell a larger share of output to the government relative to industries with a smaller procurement share, which leads to a difference-in-differences (DID) empirical strategy.

The empirical strategy is implemented with the Above-Scale Industrial Firms Panel (ASIFP) of China during 1998–2008, a period covering both pre- and post-GPL years. We supplement the data with the 2005 China Investment Climate Survey (CICS), which belongs to the World Bank Enterprise Surveys, to compute each industry's reliance on government procurement. Specifically, the treatment group consists of industries whose share of sales to the government is above the median, and the control group includes the rest industries. We also include a bunch of covariates to control for other potential determinants of the size–productivity covariance that may correlate with the treatment status at the same time.

We identify a positive and significant effect of strengthened regulation of government procurement on allocative efficiency in the data. After the implementation of the GPL, the size–productivity covariance rises in industries that sell a larger share of their output to government, compared with industries relying less on procurement. Economically, our estimate suggests that the GPL on average raises the productivity of a typical Chinese manufacturing industry by 12.49% in the post-reform period.

Our finding is robust to various concerns. We find paralleled pre-trends between the treatment and control groups, and the trends start to diverge after the implementation of GPL. Potential measurement errors in government procurement have also been carefully addressed. Our finding is consistent when using another commonly used misallocation measure, i.e., the productivity dispersion (Hsieh and Klenow, 2009). The result is also robust to the inclusion of the expectation effect,<sup>9</sup> the confounding effect from SOE procurement, restriction of the sample to domestic firms only, serial correlation in errors, and other concerns.

The economic mechanism in the model is further corroborated by a set of firm-level evidence. First, state-owned enterprises (SOEs) tend to have higher government procurement shares and lower revenue labor productivity. As SOEs are better-connected to the government by nature, this finding is consistent with our model prediction that better-connected firms sell more to the government and have lower productivity. Second, firms that devote more resources in acquiring government orders tend to have higher procurement shares and lower revenue labor productivity, as predicted by our model.

More importantly, the model generates a novel and unique prediction linking the dynamics of firms performances with the implementation of the GPL. As the GPL weakens the benefit of government connections, better-connected firms would spend less on marketing and reduce their sales to the government, which in turn decreases their total sales and raises productivity. We find supporting evidence of the prediction in the data. Proxying connected firms with SOEs and subsidized firms, we find that the sales of connected firms in industries that sell more to the government reduce after the GPL, compared with their counterparts; while the productivity of connected firms increases.

<sup>5</sup> Politically connected firms are more likely to secure government contracts. There is also evidence that firms trade political contributions for government orders. See the literature review at the end of this section for references.

<sup>6</sup> Mulabdic and Rotunno (2021) document evidence on trade protection in public procurement. In China, local protectionism as emphasized by Young (2000) could also lead to inefficient allocation of procurement orders between firms from different regions.

<sup>7</sup> Article 3 of Chapter 1 states the GPL promotes “the principles of openness and transparency, fair competition, impartiality and good faith in government procurement.”

<sup>8</sup> Physical productivity describes a firm's technological capacity, and revenue productivity pertains to profitability (Foster et al., 2008). As shown below, revenue productivity is the concept relevant to the measurement of misallocation. In the rest of the paper, productivity refers to revenue productivity when we do not make the distinction.

<sup>9</sup> The GPL was approved in June 2002 and came into effect since 2003. Governments might change their procurement practices in anticipation of the GPL during the latter half of 2002. We call this effect the expectation effect.

Our analysis marries the misallocation literature with the literature on government procurement. The misallocation literature points to the importance of misallocated production factors in explaining aggregate productivity differences, see for example, the review by Hopenhayn (2014) and Restuccia and Rogerson (2017). Such misallocation is often attributed to government policies and regulations, as detailed in Footnote 2. However, the government's role as the largest buyer in the economy has been overlooked. Our contribution to this literature is to emphasize the role the government plays as a direct market participant in allocating production factors. Closely related to our study, di Giovanni et al. (2022) also explore the government's role as a buyer through procurement and its implications on resource misallocation. But they focus on the collateral value of procurement contracts for financially constrained firms, instead of the direct effect of procurement on input allocation. Given that the misallocation of inputs is derived from sales to the government, inefficient government procurement plays a role similar to product market regulations, such as the reservation laws in India studied by Garcia-Santana and Pijoan-Mas (2014). In addition, misallocation studies often cannot directly identify particular frictions that generate misallocation (Sraer and Thesmar, 2023). Our study uses the enactment of the GPL as a quasi-natural experiment to reach causal evidence on the impact of inefficient procurement.

The literature on government procurement shows that corruption flourishes in the procurement market and that the allocation of procurement orders is often distorted. For example, politically connected firms are more likely to secure government orders (Goldman et al., 2013; Mironov and Zhuravskaya, 2016; Coviello and Gagliarducci, 2017; Baltrunaite, 2019; Schoenherr, 2019; Baranek and Titl, 2020; Brogaard et al., 2020; Brugués et al., 2022), which is one way how political connections benefit firms (Schoenherr, 2019; Brugués et al., 2022). Some studies also find that firms trade political contributions/donations for government orders (Baltrunaite, 2019; Titl and Geys, 2019), as we model in this paper.<sup>10</sup> Compared to this literature, our study links misallocation of procurement orders to misallocation of production factor and quantifies the productivity effect of inefficient procurement, thus bridges the gap between this literature and the macro misallocation literature. In this regard, Brugués et al. (2022) is closest to our study. The two papers differ in three aspects. First, Brugués et al. (2022) evaluate the welfare cost of misallocated procurement orders, while we focus on production efficiency as in Hsieh and Klenow (2009). Second, Brugués et al. (2022) examine differences in revenue productivity across firms, while we study the size-productivity covariance following Bartelsman et al. (2013). Third, Brugués et al. (2022) study the impact of changes in firm's political connections on resource allocation in Ecuador, while we study the impact of a law regulating government procurement in China.

Studies in the government procurement literature have also demonstrated the impact of changes in legislation or procurement practices on procurement outcomes (Bandiera et al., 2009; Coviello and Mariniello, 2014; Neu et al., 2015; Baltrunaite, 2019; Bosio et al., 2022). In this regard, we provide an evaluation of China's reform in government procurement towards a more transparent and standardized procurement process, which covers more ground of procurement than most of the laws and legislations studied in the existing literature. Standardization and transparency in the procurement process are crucial not only for China but also for many other countries as they are critical requirements of the WTO Agreement on Government Procurement. We show both theoretically and empirically that such a reform is most successful when misallocation of procurement orders is more severe, which has implications for developing countries considering similar reforms.

Our research also relates to a new but burgeoning literature studying how government procurement orders affect firm performance, such as access to credit (Hebous and Zimmermann, 2021; di Giovanni et al., 2022), investment (Cohen and Malloy, 2016; Hebous and Zimmermann, 2021), innovation (Slavtchev and Wiederhold, 2016; Czarnitzki et al., 2018), employment (Gugler et al., 2020), and firm survival and growth (Ferraz et al., 2015; Cappelletti and Giuffrida, 2021; Lee, 2021). This literature mainly investigates the effects of procurement orders on *individual firms*. By contrast, we focus on the impact of misallocated procurement orders on resource allocation *between firms*.

Finally, a large literature has studied resource misallocation in China due to factor or product market distortions (Brandt et al., 2013; Cheremukhin et al., 2015; Adamopoulos et al., 2022; Tombe and Zhu, 2019). One prominent example of such distortions is the benefits that SOEs receive in the capital market (Song et al., 2011; Hsieh and Song, 2015). Our paper emphasizes the role of government procurement in driving product market distortions, much like the role of local protectionism emphasized in Young (2000).

The rest of the paper is organized as follows. Section 2 describes the GPL and government procurement in China in detail. Section 3 presents the theoretical model that guides our empirical analysis. Section 4 discusses the data and develops the empirical strategy. Section 5 presents the empirical results. Section 6 concludes.

## 2. Background

Modern government procurement using a tendering and bidding system only began in China in the 1980s. To increase competition among SOEs, the State Council created several localized tendering and bidding markets throughout China that introduced market mechanisms in the procurement of construction and public works. These measures increased efficiency in allocating capital, although government procurement was still severely under-regulated in the 1990s. Local protectionism and corruption were pervasive in the government procurement markets. The use of government funds was inefficient (USCBC, 2012).

<sup>10</sup> These studies in general find that corruption lowers the efficiency of the procurement process, in contrast to an "efficient grease" view that corruption mitigates market frictions or helps firms to get around excess bureaucracy and regulation (Schoenherr, 2019; Brugués et al., 2022).

In response to these challenges, on January 1, 2003, China implemented the *Government Procurement Law of the People's Republic of China*. The GPL provides the first national comprehensive framework for procurement across all levels of government.<sup>11</sup> The GPL was part of a broader set of economic reforms aimed at strengthening China's market-oriented economic system and promoting competition, with a particular focus on the government procurement market. It was also seen as an important step in meeting the requirements of the WTO Agreement on Government Procurement during the rounds of negotiations leading up to China's entry into the WTO.

The GPL tries to standardize the procurement process following international practices, under the principles of openness and transparency, fair competition, impartiality and, good faith (Article 3, Chapter 1). In particular, the GPL defines the parties involved in public procurements as different levels of government (Chapter 2), and mandates that public tendering shall be the principal method of government (Article 26, Chapter 3), formalizes the procedures of procurement (Chapter 4), and emphasizes the use of formal contracts (Chapter 5). Suppliers are allowed to submit inquiries about the procurement and appeal to the supervising government agency or the court upon the announcement of the procurement decision (Chapter 6). Procurement should also be subject to supervision and inspection on a regular basis (Chapter 7). These measures of standardization aim at leveling the playing field for all firms during every stage of government procurement and reducing the incidence of misallocated procurement contracts.

The GPL has been effectively implemented in practice since its enactment. On August 5, 2003, the Ministry of Finance released the *Opinions on Pushing Forward the Reform of Government Procurement System in an All-Round Way*, stating “in order to ensure the successful implementation of the Government Procurement Law, urge the governmental departments in all regions to further reform the government procurement system, and standardize the government procurement procedure.” On August 18, 2003, the General Office of the State Council distributed this document to all levels of government, ministries and commissions of the State Council, and all departments directly under the State Council, and asked them to carry it out accordingly.<sup>12</sup> According to a report by the Treasury Department of the Ministry of Finance in 2003, the Ministry of Finance further formulated 13 detailed administrative measures and operational regulations, and formed a government procurement contract template writing group of experts and scholars to come up with major contract templates. The report called 2003 the “Year of Institutional Construction” for government procurement.<sup>13</sup> The GPL has been regulating government procurement until it was first revised in 2014. As 2014 is beyond the scope of the data in this study, the amendment is not discussed.

The GPL greatly changed government procurement practice in China. According to reports of the Ministry of Finance, the enactment of the GPL led to the establishment of an open and transparent procurement market, which helped the government save a large scale of procurement funds.<sup>14</sup> In 2005, the government in total saved 32.5 billion RMB out of more than 250 billion RMB in government procurement.<sup>15</sup> During 2003–2012, ten years since the enactment of the GPL, the governments in total saved 660 billion RMB procurement funds, accounting for around 11% of the total budget.<sup>16</sup> These estimates indicate significantly improved efficiency in using procurement funds.

Our paper studies the resource allocation implication of improved efficiency in procurement due to the GPL. As the GPL covers all stages of procurement, it begs the question of how should we understand the GPL in terms of economic logic. Misallocation of procurement orders indicates that the allocation process depends on non-market factors other than the price and quality of products. Our view is that the standardization of the procurement process does not directly eliminate such non-market factors. By instituting clear guidelines and transparent procedures for procurement, the GPL mainly reduces the advantages enjoyed by firms favored by the government, leveling the playing field for all suppliers bidding for government contracts. As a result, it fosters fair competition and a more efficient allocation of production factors among firms. The next section proposes a simple model that captures these ideas.

### 3. Model

In this section, we present a simple model to show how the allocation of government procurement orders affects the allocation of production factors between firms. The model leads to theoretical predictions that are tested in the rest of the paper.

#### 3.1. Model setup

Consider a one-sector economy. A single final good is produced using  $I$  differentiated intermediate goods. A representative firm produces the final good with a CES technology,

<sup>11</sup> Along with the Tendering and Bidding Law enacted in 2000, the GPL serves as the primary legislation governing the Chinese procurement market. While the Tendering and Bidding Law mainly regulates procurement by SOEs and large-scale public works projects, the GPL regulates government procurement.

<sup>12</sup> See the *Circular of the General Office of the State Council on Distributing the Opinions of the Ministry of Finance on Pushing Forward the Reform of Government Procurement System in an All-Round Way*.

<sup>13</sup> The Treasury Department of the Ministry of Finance of China, 2003, “The Reform of the Government Procurement System Has Reached a New Level: The Implementation of The 2003 Government Procurement Law Has Achieved Initial Results.” *China Government Procurement*, 12, 4–9.

<sup>14</sup> The saving in procurement funds is accompanied by a steady increase in the size of government procurement in China. See the July 9, 2012 report “Ministry of Finance: Ten-fold Growth of Government Procurement Market Does Not Conflict with Statistics for Cuts in Expenditure,” which can be accessed at [http://www.gov.cn/jrzq/2012-07/09/content\\_2178960.htm](http://www.gov.cn/jrzq/2012-07/09/content_2178960.htm).

<sup>15</sup> [http://www.ccg.gov.cn/ldjh/201711/t20171108\\_9124432.htm](http://www.ccg.gov.cn/ldjh/201711/t20171108_9124432.htm).

<sup>16</sup> [http://www.gov.cn/jrzq/2012-07/09/content\\_2178960.htm](http://www.gov.cn/jrzq/2012-07/09/content_2178960.htm).

$$Q = \left( \sum_i Q_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}},$$

where  $i$  indexes the differentiated good. Each differentiated good is produced by a single firm, such that we also use  $i$  to index the firm. Let the final good be the numeraire; the inverse market demand for an intermediate good is given by  $P_i = Q^{\frac{1}{\sigma}} Q_i^{-\frac{1}{\sigma}}$ .

Demand for the differentiated good also comes from the government. The government provides a public service which is produced using the differentiated goods according to the same CES technology as market production,  $G = \left( \sum_i G_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$ , where  $G_i$  is the use of the intermediate good by the government. The government has a fixed budget  $E$  that is used for purchase of intermediate goods. Given the budget, the government maximizes the following objective:

$$\max_{G_i} \left( \sum_i \gamma_i G_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}.$$

Note that the government puts different weights on differentiated goods than that in the production function. This is motivated by the fact that the government may have other objectives than merely maximizing the public services. For example, the government may also seek to provide implicit subsidies to local firms. It can also be due to corruption in the procurement process: the government officials in charge of procurement can benefit personally from misallocated production orders. We take a reduced-form approach to model the inefficiencies in the procurement process. The effect of any such inefficiencies on the allocation of procurement orders is captured by the weights in the objective function. As the quantity and quality of public services are often hard to measure, whereas government budget is more measurable, we believe it is reasonable to allow the government to misplace orders between firms but not to alter the budget.<sup>17</sup> Given the fixed budget and the government's objective, the demand for differentiated good is given by  $G_i = \frac{\gamma_i^\sigma \bar{P}^{1-\sigma}}{\sum_j \gamma_j^\sigma \bar{P}_j^{1-\sigma}} E$ , where we use  $\bar{P}_i$  to allow for the possibility that a firm may sell to the government at a different price than the market price.

The intermediate good firms pay a fixed cost in terms of labor  $f$  to produce. Once the fixed cost is paid, the firm has access to a linear technology,

$$Q_i + G_i = A_i L_i^y,$$

where the left hand side says that a firm's output are sold to the government and in the market,  $A_i$  is the firm's physical productivity, and  $L_i^y$  is labor input used in production. The demand from the market and the government are independent. The firm maximizes profit from selling in the market taking the demand as given, which leads to the pricing rule  $P_i = \frac{\sigma}{\sigma-1} \frac{W}{A_i}$  with  $W$  being the wage rate.

The quantity sold in the market is then  $Q_i = Q \left( \frac{\sigma}{\sigma-1} \frac{W}{A_i} \right)^{-\sigma}$ .

Different from selling in the market, firms selling to the government can manipulate  $\gamma_i$  at a cost. In particular, the labor required to achieve  $\gamma_i$  is given by,

$$L_i^m = \eta_i \gamma_i^\theta A_i^{\sigma-1}, \quad (1)$$

where  $\eta_i$  indicates a firm's inherent connection with the government, and the scaling by  $A_i^{\sigma-1}$  makes sure that the cost of manipulating  $\gamma_i$  is proportional to sales. We impose  $\theta > \sigma$  such that the profit maximization problem introduced below is well defined. Since the cost function is key to our model, a discussion is in order. In some sense, we can view  $L_i^m$  as a marketing cost. We will call it the marketing cost in this paper, but it is different from the usual marketing cost of selling in the market which we do not model here. It is the cost beyond the usual marketing cost that is required to acquire government orders. Because of the government's fixed budget, selling to the government creates profit which can be viewed as rents. One way to think about the cost is it reflects the firm's rent-seeking activities. Labor used for such unproductive rent-seeking activities represents dissipated rents. As we will show below, the cost function then results in a fixed fraction of the rents dissipated due to rent-seeking. Alternatively, if the cost of manipulating  $\gamma_i$  is paid to the government or government officials, the cost determines the way of sharing the rents between the government and firms.<sup>18</sup> The rents paid to the government however is not monetary, but in the form of labor input by the firm. One example for such kind of rent sharing is that the government often demands SOEs to take on responsibilities other than profit maximization (Bai et al., 2000, 2006). The cost function then says that the firms and the government adopt a fixed fraction sharing rule, if we were to ignore the differences in government connections. This alternative view is consistent with the complicated relationship between government and firms in China (Bai et al., 2020).

<sup>17</sup> This is in line with the GPL. Article 6, Chapter 1 of the GPL states that "Government procurement shall be conducted strictly in accordance with the budget approved."

<sup>18</sup> We could imagine that this scenario arises in an explicit model of the procurement process in which government and the firm bargain to split the rent. Some readers might want to view the cost as firms trade political contributions or donations for government orders, as found in Baltrunaite (2019) and Tittl and Geys (2019). Our model also has that flavor. But we do not explicitly model the bargaining process or use political contributions or donations to describe the cost because in China, the cost might manifest itself in complicated forms (Bai et al., 2020). For our purpose, it does not matter where the cost goes.

### 3.2. Model analysis

We focus on how selling to the government distorts resource allocation in this subsection. The profit maximization problem of selling to the government is given by,

$$\max_{\tilde{P}_i, \gamma_i} \left( \tilde{P}_i - \frac{W}{A_i} \right) G_i - \eta_i \gamma_i^\theta A_i^{\sigma-1} W.$$

Solving the problem implies the same constant markup pricing as above,  $\tilde{P}_i = \frac{\sigma}{\sigma-1} \frac{W}{A_i}$ . The optimal choice of  $\gamma_i$  is given by,

$$\gamma_i = \left( \frac{\sigma \Phi}{\theta \eta_i} \right)^{\frac{1}{\theta-\sigma}}, \quad (2)$$

where  $\Phi = \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \frac{E}{W^\sigma \sum_j \gamma_j^\theta \tilde{P}_j^{1-\sigma}}$  summarizes the aggregate market condition for selling to the government. Equation (2) shows the sources of resource misallocation:  $\eta_i$ . Firms otherwise would choose the same marketing effort if not for the differences in their connections to the government. If such differences are removed, each firm's sales to the government would be solely determined by productivity. The allocation of government orders across firms would be the same as the market outcome. The use of intermediate inputs in producing public services would be efficient, given the set of firms that enters the market.<sup>19</sup> This is one type of resource misallocation in this model. On the other hand, the above discussion shows that the marketing cost can be either viewed as dissipated rents or resources that are steered by the government to other (possibly unproductive) uses. Therefore, there is also resource misallocation between productive and unproductive activities (Baumol, 1990; Auriol et al., 2016). We however do not pay particular attention to this type of misallocation. We focus on the first type of misallocation, as in the recent literature.

Our first set of results describes how government connections affect firm's marketing activity and sales to the government. These results are summarized in the following proposition.<sup>20</sup>

**Proposition 1.** *Firms better-connected to the government (smaller  $\eta_i$ ) devote more resources to marketing (larger  $L_i^m$ ), sell more to the government (larger  $G_i$ ), and have higher shares of sales to the government (larger  $\frac{\tilde{P}_i G_i}{P_i Q_i + \tilde{P}_i G_i}$ ), conditional on physical productivity.*

Lower marketing cost of connected firms induce them to raise the marketing input, which increases sales to the government and the share of sales to government. This proposition predicts several cross-sectional correlations between firm-level variables that we later test in the data.

To derive a measure of resource misallocation, we look at firm's revenue productivity following the literature. Since labor is the only input, revenue productivity in our model is defined as,

$$LP_i = \frac{P_i Q_i + \tilde{P}_i G_i}{L_i^y + L_i^m + f}.$$

We can separately look at the production for the market and for the government. Ignoring the fixed cost, let  $L_i^{yQ}$  be the labor used in production for the market, and  $L_i^{yG}$  be that for the government including the marketing cost. We can rewrite the productivity as,

$$LP_i = \left( \frac{P_i Q_i}{L_i^{yQ}} \frac{L_i^{yQ}}{L_i^{yQ} + L_i^{yG}} + \frac{\tilde{P}_i G_i}{L_i^{yG}} \frac{L_i^{yG}}{L_i^{yQ} + L_i^{yG}} \right) \frac{L_i^{yQ} + L_i^{yG}}{L_i^y + L_i^m + f}.$$

It is thus a weighted average of productivity in production for the government and that for the market, adjusted by the ratio of production labor to total labor input. From the firm's pricing strategy, productivity for the market is  $\frac{\sigma}{\sigma-1} W$ , and that for the government is  $\frac{\sigma}{(\sigma-1)+1/\theta} W$ . Selling to the government has lower productivity. Therefore, given total employment, a firm that uses more labor to produce for the government has lower productivity. Given the allocation of labor between producing for the government and that for the market, a larger firm have higher productivity due to the fixed cost. In the data, we are not able to tell whether a worker produces for the government or for the market. We thus further rewrite productivity as follows,

$$LP_i = \frac{W}{\frac{P_i Q_i}{P_i Q_i + \tilde{P}_i G_i} \frac{\sigma-1}{\sigma} + \frac{\tilde{P}_i G_i}{P_i Q_i + \tilde{P}_i G_i} \frac{(\sigma-1)+1/\theta}{\sigma} + \frac{fW}{P_i Q_i + \tilde{P}_i G_i}} \quad (3)$$

This equation links a firm's productivity to its share of sales to the government instead of share of employment. The next proposition follows naturally.

<sup>19</sup> We ignore the firm entry problem in this paper because we only analyze existing firms in our empirical exercise. In the model, we simply hold the number of firms serving the market as given.

<sup>20</sup> The proof of the proposition is as follows. Equations (1) and (2) together imply that  $L_i^m \propto \eta_i^{-\frac{\sigma}{\theta-\sigma}} A_i^{\sigma-1}$ , namely better-connected firms spend more on marketing, conditional on physical productivity. These firms have higher sales to the government due to higher  $\gamma_i$ . As market demand depends only on physical productivity, they also have higher shares of sales to the government.

**Proposition 2.** A firm's productivity is decreasing in its share of sales to the government, conditional on total sales.

Propositions 1 and 2 together imply the following corollary:

**Corollary 1.** Firms better-connected to the government (smaller  $\eta_i$ ) devote more resources to marketing (larger  $L_i^m$ ), and have lower productivity, conditional on firm sales.

These predictions will also be tested later in our data. The effect of government connections on productivity turns out to be the key to understand our main empirical prediction below.

As discussed above, the share of sales to the government in our model is determined only by  $\eta_i$ . Therefore, in an efficient allocation, ignoring the fixed cost, productivity in production ( $\frac{P_i Q_i}{L_i^{yQ}} \frac{L_i^{yQ}}{L_i^{yQ} + L_i^{yG}} + \frac{\bar{P}_i G_i}{L_i^{yG}} \frac{L_i^{yG}}{L_i^{yQ} + L_i^{yG}}$ , or relatedly  $\frac{P_i Q_i}{P_i Q_i + \bar{P}_i G_i} \frac{\sigma-1}{\sigma} + \frac{\bar{P}_i G_i}{P_i Q_i + \bar{P}_i G_i} \frac{(\sigma-1)+1/\theta}{\sigma}$ ) would be the same for all firms. In this case, larger firms will have higher productivity than smaller firms because the fixed cost is spread over larger sales. Efficient allocation thus implies a positive covariance between productivity and firm size in our model. Bartelsman et al. (2013) show that distortions can lower the size–productivity covariance, which can be viewed a measure of allocative efficiency. Following them, we also use the size–productivity covariance as our measure of resource misallocation. To see why misallocation lowers the covariance, consider the following case. Suppose all firms have the same productivity, such that the difference in firm size is driven only by government connections. A firm achieves higher sales only through higher sales to the government, which lowers the firm's productivity. As long as the fixed cost is small compared to sales, the size–productivity covariance will be negative.<sup>21</sup> In reality, firms differ in both physical productivity and government connections, so the covariance would be lower if the role of government connections becomes larger. The covariance would also be lower if the size of government procurement becomes larger relative to that of market demand.

### 3.3. The impact of the GPL

We now discuss the impact of the GPL on aggregate outcomes within the model. To conduct this analysis, we need to establish a connection between the GPL and model parameters.

We interpret the enactment of the GPL as an increase in the cost elasticity parameter, denoted as  $\theta$ . As discussed in Section 2, the GPL was implemented in response to the inefficient utilization of public funds in government procurement. Its primary objective revolves around the establishment of an open and transparent procurement market with standardized procedures. Consequently, the law does not directly aim to eliminate government connections. However, by standardizing the procurement process, it introduces greater difficulty for firms to leverage government connections for securing government orders. Specifically, this should reduce the advantage that well-connected firms have over their less-connected counterparts in obtaining procurement orders. This effect can be accurately represented by an increase in the parameter  $\theta$ .<sup>22</sup>

An increase in  $\theta$  has two effects. First, firms allocate fewer resources to marketing for government orders. Second, the allocation of procurement orders becomes more dependent on productivity. Both of these effects contribute to an overall increase in aggregate productivity. To express this formally, given that the final good is identical for both the market and the government, we can calculate aggregate labor productivity in the model using the following equation:

$$LP = \frac{Q+G}{L} = \left( \sum_i A_i^{\sigma-1} \right)^{1/(\sigma-1)} \frac{L - L^{yG} - L^M - If}{L} + \frac{\left( \sum_i \eta_i^{\frac{\sigma-1}{\sigma-\theta}} A_i^{\sigma-1} \right)^{\frac{\sigma}{\sigma-1}}}{\sum_i \eta_i^{\frac{\sigma}{\sigma-\theta}} A_i^{\sigma-1}} \frac{L^{yG}}{L}$$

where  $L^{yG} = \frac{E}{\left( \sum_i A_i^{\sigma-1} \right)^{1/(\sigma-1)}}$  is total labor required for producing output selling to the government,  $L^M = \frac{1}{\theta(\sigma-1)} \frac{E}{\left( \sum_i A_i^{\sigma-1} \right)^{1/(\sigma-1)}}$  is total

labor required for marketing,  $\left( \sum_i A_i^{\sigma-1} \right)^{1/(\sigma-1)}$  is the aggregate labor productivity for market production, and  $\frac{\left( \sum_i \eta_i^{\frac{\sigma-1}{\sigma-\theta}} A_i^{\sigma-1} \right)^{\frac{\sigma}{\sigma-1}}}{\sum_i \eta_i^{\frac{\sigma}{\sigma-\theta}} A_i^{\sigma-1}}$  is

the aggregate labor productivity for the government service. Aggregate productivity is simply a weighted average of productivity in the two segments: market and government. The weights do not add up to one due to the presence of fixed cost and marketing cost. Evidently, an increase in the parameter  $\theta$  reduces marketing expenditures, freeing up more resources for market production and consequently raising aggregate productivity.  $\eta_i$  induces a wedge in supplying to the government. Its impact is smaller with a higher  $\theta$ , leading to a higher labor productivity.

<sup>21</sup> To be accurate, raising sales due to better government connections lowers productivity if  $f < \frac{P_i Q_i + \bar{P}_i G_i}{\theta \sigma W}$ .

<sup>22</sup> People might argue that the GPL might shift the marketing cost function up for all firms. The cost function might shift up, but a parallel shift is not the right way to think about the GPL. As the allocation of the procurement orders depends on the relative size of  $\gamma_i$ , an upward shift of the cost function simply makes all firms reduce their desired  $\gamma_i$  without affecting the relative standing of firms in receiving government orders. As the cost function is a power function (essentially a Cobb–Douglas setup), the reduction in  $\gamma_i$  and the increase in the cost function would offset each other such that the firm's share of marketing cost in government sales is constant. In this sense, it does not matter if the GPL shifts the marketing cost function.

**Table 1**  
Calibration of Key Model Parameters.

| Model Parameter     | Interpretation                                                               | Source or Calibration Target                                                                 |
|---------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| $\delta = 6$        | Elasticity of substitution between varieties                                 | Broda and Weinstein (2006)                                                                   |
| $\alpha = 6$        | Shape parameter of the Pareto Productivity draw                              | Share of top 5% firms in private sales is 68% in the Chinese sample                          |
| $\underline{A} = 1$ | Scale parameter of the Pareto Productivity draw                              | Normalization                                                                                |
| $L = 1,000,000$     | Total labor supply                                                           | Average firm size normalized at 1 with 1,000,000 simulated firms                             |
| $f = 2/27$          | Fixed cost                                                                   | Share of economic profit is 10% in U.S.                                                      |
| $\rho = 0.1881$     | The impact of productivity on the probability of receiving government orders | Probit regression coefficient of receiving government orders on (log) market sales is 0.0392 |
| $\phi = 0.7178$     | The impact of productivity on government connection                          | Regression coefficient of (log) government sales on (log) market sales is 0.7178             |
| $\theta = 8$        | The cost elasticity of acquiring government orders                           | Share of entertainment and travel costs expenditures is 1.4% of total sales                  |

Note: Government expenditure is calibrated to the share of government expenditure in total sales. We experiment with three shares: 0.01, 0.05, and 0.1.

Although the concept is easy to understand, we however are not able to compute the above labor productivity in data, as we can observe neither  $G$  nor  $G_i$  in data. Since the increase in aggregate productivity is driven by better allocation of resources, we argue that it should manifest in changes in the size-productivity covariance, which serves as a measure of allocative efficiency. There are two reasons. First, a decrease in marketing costs reduces the productivity differences between production for the government and production for the market.<sup>23</sup> In the extreme case of  $\theta$  approaching infinite, government orders are allocated solely based on productivity, and the economy achieves efficiency in government procurement. Productivity in producing for the government is then the same as that for the market. The size-productivity covariance is the same as that under market allocation. Second, firms with better connections narrow their advantages in selling to the government, reducing the difference in the share of sales to the government. Both effects work towards raising the size-productivity covariance. The effects are stronger in industries that rely more on government sales.

It is not possible to analytically show how an increase in  $\theta$  changes the size-productivity covariance and its relationship with aggregate productivity. We thus supplement the above analysis with a numerical example. For this purpose, we calibrate the model and perform counterfactual analysis with the calibrated model.

We first describe our calibration strategy. We set the elasticity of substitution  $\sigma$  to be 6, which is roughly in the middle of the estimates in Broda and Weinstein (2006). The literature has found that firm size often follows a Pareto distribution in the upper tail (Gabaix, 2016). In our model, firm size distribution is closely linked to the distribution of firm productivity. We thus assume firm productivity is drawn from a Pareto distribution.<sup>24</sup> The shape parameter of the Pareto distribution (denoted  $\alpha$ ) is calibrated to the share of top 5% firms in total private sales in our Chinese data sample introduced below, which gives a value of 4.6.<sup>25</sup> The scale parameter of the Pareto distribution (denoted  $\underline{A}$ ) does not affect our analysis in any way, we thus normalize it at 1. We normalize average firm employment to be 1, such that when we draw 1,000,000 firms in our simulation, total labor input is also set to be 1,000,000. Given the average firm size, the fixed cost is set to be  $2/27$  such that the share of economic profit in a distortionless economy is 10%, as in the U.S. (Basu, 2019).

The rest of the parameters are calibrated to the Chinese data. A key to our calibration is the distribution of government connections  $\eta_i$ . For an average industry with sales to the government, the share of firms selling to the government is 20%. So we assume only 20% of firms have finite  $\eta_i$  in our model. The set of firms are selected according to the following index,

$$\text{index}_i = \rho \log A_i + \epsilon_i,$$

where  $\epsilon_i$  is a random draw from a standard normal distribution. Firms with a higher index are able to sell to the government, and  $\rho$  determines the correlation between receiving government orders and firm productivity. In the data, firm productivity is closely related to market sales. We thus calibrate  $\rho$  to match the regression coefficient in a probit regression of whether selling to the government on the log of market sales. The coefficient in the data is 0.0392, leading to a value of 0.1881 for  $\rho$ . Similarly, we assume  $\eta_i$  for firms with positive government sales is related to firm productivity as follows,

$$\log \eta_i = \phi \log A_i + \epsilon_i,$$

where  $\epsilon_i$  again is a random draw from a standard normal distribution. We calibrate  $\phi = 0.6116$  to match the regression coefficient of the log of government sales on the log of market sales in the data, 0.7178. Table 1 summarizes our calibration results.

<sup>23</sup> Remember that the productivity in production for the government is  $\frac{\sigma}{(\sigma-1)+1/\theta}W$  and that for the market is  $\frac{\sigma}{(\sigma-1)}W$ .

<sup>24</sup> Given the fixed cost and distortions in government procurement, firm size in our model will not be exactly Pareto. But for large firms, these effects should be small.

<sup>25</sup> The Pareto shape parameter of the distribution of private sales is 1.1, very similar to findings in other studies of firm size distribution. So the calibration of this parameter is not confined to the Chinese data.

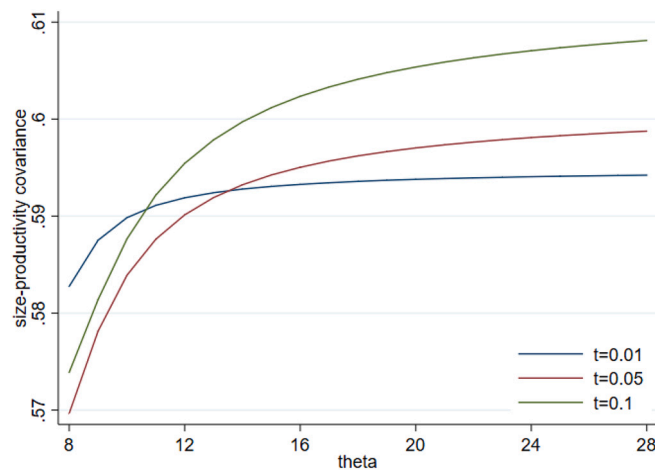


Fig. 1. The Relationship Between  $\theta$  and the Size-Productivity Covariance in the Model.

Finally, we have set the size of total government procurement  $E$  and the cost elasticity of acquiring government orders  $\theta$ , the two variables closely linked to our discussion of the effect of the GPL. We set  $E$  to match the share of government procurement in total output. We experiment with three values: 0.01, 0.05, and 0.1, which roughly correspond to the range of shares for different industries in the data.  $\theta$  is set at 8 in our baseline, such that the ratio of marketing costs to government sales is 1/48. Cai et al. (2011) argue that the share of entertainment and travel costs expenditures can be used as a proxy for spending on government relations. In the data, this share is 1.4%. If the marketing costs for government orders are part of these expenditures, the value we assign to  $\theta$  seems to be reasonable because it suggests the spending ratio for government sales is larger than that for market sales. We gradually increase  $\theta$  to 28 to see how the size-productivity covariance changes as  $\theta$  increases.

Fig. 1 presents our simulation results regarding changes in the size-productivity covariance. Two results immediately emerge. First, the size-productivity covariance increases as  $\theta$  increases. Second, the increase is larger when the share of government procurement in total output is larger. This confirms our analysis above. We summarize them in the following main prediction that we test in the paper.

**Prediction 1.** *Tightened regulation of government procurement due to the GPL raised the size-productivity covariance in industries that rely more on government procurement relative to that in industries rely less on procurement.*

Besides the main prediction, Fig. 1 provides more interesting details worth mentioning. The increase in the size-productivity covariance is fast starting from the baseline of  $\theta = 8$ , but quickly slows down when  $\theta$  goes above 12. It suggests that initial improvement of the procurement process from a low level can be very effective, but further improvement might be less so. Developing countries are more likely to reap gains from reform on government procurement. The relationship between government procurement share and the size-productivity covariance is not monotonic. In particular, when  $\theta$  is high enough, a higher government procurement share raises the size-productivity covariance. This seems counter-intuitive. This is mainly because we only allow 20% firms to sell to the government, such that the misallocation between firms selling and not selling to the government can not be reduced by a higher  $\theta$ .<sup>26</sup> For  $\theta$  higher enough, selling to the government does not have lower labor productivity. The size-productivity covariance is then determined by the firm size distribution. The covariance is raised by a larger procurement share because of a shift of the firm size distribution to the right. In this case, the effect of a rise in government procurement share can be viewed as a shift in the underlying firm productivity distribution. Size-productivity covariance no longer serves as a good proxy for misallocation in this cross-sectional comparison of levels. This result however does not affect our empirical prediction because it compares changes.

Fig. 2 further shows that labor productivity is positively related to the size-productivity covariance for the case where the share of government expenditure is 0.05. The impact of the GPL on the size-productivity covariance translates into an impact on aggregate productivity. Our study on the size-productivity covariance thus sheds light on the impact of the GPL on productivity, which is hard to measure in data.

#### 4. Empirical strategy

This section discusses the empirical strategy. We first present the specification of the empirical model, then the data, and finally the construction of key variables.

<sup>26</sup> Our second point in the verbal discussion above, “firms with better connections narrow their advantages in selling to the government, reducing the difference in the share of sales to the government,” no longer applies in this case.

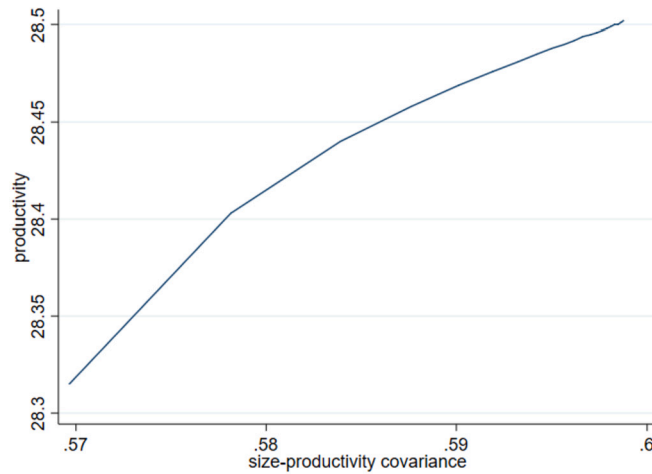


Fig. 2. The Relationship Between Aggregate Productivity and the Size-Productivity Covariance in the Model ( $t=0.05$ ).

#### 4.1. Empirical specification

The prediction from our theoretical analysis can be tested by examining changes in allocative efficiency before and after the GPL. However, a simple comparison of the pre- and after-GPL periods for industries with government procurement cannot allow us to isolate the effect of the GPL. To establish a causal relationship between the GPL and resource allocation, we adopt a DID estimation strategy. The treatment group includes industries with the share of sales to the government higher than the median, and the rest belong to the control group. The identification assumption is that the extent of reliance on procurement is an industry characteristic that is determined by the type of public services the government provides, but is not shaped by the procurement practice itself. We later provide supporting evidence for this assumption, and in sub-section 4.3, we discuss how we measure the treatment status in data.

The baseline econometric specification for the DID analysis is given as follows:

$$y_{it} = \beta \cdot gs_i \times post_t + \mathbf{X}_{it}'\gamma + \lambda_i + \lambda_t + \epsilon_{it}, \quad (4)$$

where  $i$  indexes a four-digit industry and  $t$  indexes a year.  $y_{it}$  is the measure of resource misallocation in industry  $i$  of year  $t$ ;  $post_t$  is a dummy indicating the enactment of the GPL, which equals 1 for years after 2002, and 0 otherwise;  $gs_i$  is an indicator for the treatment status,  $gs_i = 1$  if an industry  $i$  is in the treatment group, and  $gs_i = 0$  if in the control group.  $\lambda_i$  and  $\lambda_t$  are industry- and year-fixed effects that control for time-invariant industry characteristics and general time trends. Many factors are behind the observed size-productivity covariance and these factors could also correlate with our treatment and confound our estimates. We have controlled in  $\mathbf{X}_{it}$  a set of such variables, which is explained in detail later. These controls reflect our best effort to control other determinants of resource allocation. Although it is not possible to exhaust all potential determinants, our working assumption is that the omitted determinants are not likely to correlate with the industry characteristics that shape the treatment status. Besides trying to control the potential determinants of the size-productivity covariance, we also use an alternative misallocation measure, the productivity dispersion (Hsieh and Klenow, 2009), to check the robustness. Following Bertrand et al. (2004), we cluster the errors  $\epsilon_{it}$  at the industry level.

We are interested in the estimate of  $\beta$ . According to our discussion above, industries relying more on government procurement are affected more by the GPL. Resource misallocation in these industries should improve compared with that in industries in the control group. Therefore, we expect a positive estimate of  $\beta$ .

#### 4.2. Data

We implement the empirical model using a dataset with detailed firm production data, supplemented by a few other data sources.

The main dataset that we use is the Above-Scale Industrial Firms Panel (ASIFP) database maintained by China's National Bureau of Statistics (NBS) during 1998–2007. The ASIFP database includes all SOEs, and non-SOEs with annual sales above 5 million RMB in mining, manufacturing, and utilities industries. We drop all non-manufacturing firms to focus on the manufacturing sector. The database contains detailed balanced sheet information on individual firms, which allows us to construct measures of resource allocation and some key control variables for our estimation. The database has been widely used in the literature. We clean the ASIFP database following standard procedures in Cai and Liu (2009) and Brandt et al. (2012).

We supplement the ASIFP with the China 2005 Investment Climate Survey (CICS), which belongs to the World Bank Enterprise Surveys. This survey covers 12,400 firms located in 120 cities. The surveyed firms span all two-digit manufacturing industries and all Chinese provinces except Tibet. In each province, the survey includes the provincial capital city and additional cities selected according to their relative economic size within the province. In Shanghai, Tianjin, Beijing, and Chongqing, the four centrally-

administered municipalities, 200 firms are surveyed. In all other cities, 100 firms are sampled. The survey includes both the basic information of surveyed firm, such as its industry and city, and firms' financial information such as total sales, employment, and so on. Most firms report information in 2004, whereas some of them date back to 2002. We use the CICS data to compute each industry's share of sales to government, which is the basis for determining the treatment status.

We also use tariff data from the World Integrated Trade Solutions (WITS) to construct key control variables, and the 2003 CICS data and the 2002 Input–Output (IO) Table of China for alternative measures of government procurement share. We match all these datasets together to construct a panel at four-digit industry level during 1998–2007.

#### 4.3. Variable construction

We next discuss the construction of the treatment status and the dependent variable for our estimation. The construction of other industry-level controls is straightforward.

**Treatment Status.** We use the CICS data to determine the treatment status. In particular, we utilize information from the question “what percent of your products are sold to the government in 2004” with firm's total sales to compute sales to government for each firm. We then aggregate total sales and sales to government to two-digit manufacturing industries to compute each industry's share of sales to government. Treatment status for two-digit industries is then determined by whether the share is above or below the median. We match it to four-digit industries using the China Industry Classification Standard GB/T 4754-2002. Evenly dividing the two-digit industries into treatment and control groups using the median of procurement share as a cut-off ensures the two groups are also roughly balanced at the four-digit level.

The treatment and control groups drastically differ in reliance on government procurement. The average government procurement share in the control group is 0.77%, while that in the treatment group is roughly 4 times larger at 3.04%. The industry with the highest government procurement share is the transport equipment manufacturing industry, which is closely related to the pervasive use of vehicles in governments; the manufacturing industry of clothing, footwear, and apparel, which is probably due to the large demand for uniforms from the government; and manufacturing industry of instruments, meters, and office appliances, which could be driven by the demand for office supplies such as printers. On the contrary, industries with the lowest government procurement shares are recycling and disposal of waste.<sup>27</sup> Industries in the treatment and control groups conform with our prior about the pattern of government procurement. The procurement share in each industry seems to depend on the characteristics of government operation, rather than the procurement practice. Therefore, we believe that the indicators constructed based on the CICS can reflect the intensity of government procurement in various industries.

**Resource Misallocation.** We use the firm size–productivity covariance as our measure of resource misallocation, following Bartelsman et al. (2013). The covariance comes from the Olley and Pakes (1996) decomposition of an industry-level productivity index, defined as the weighted average of (log of) firm-level productivity, into an unweighted firm-level average and a covariance term. Bartelsman et al. (2013) argue in theory and in data that the covariance is a robust measure of allocative efficiency. Our theoretical analysis confirms its effectiveness as a measure of misallocation in a model where firm-level idiosyncratic policy distortions are driven by government procurement.

Formally, the Olley and Pakes (1996) decomposition is given by

$$\Omega_{it} = \sum_{f \in I_i} \theta_{it} \omega_{ft} = \bar{\omega}_{it} + \sum_{f \in I_i} (\theta_{ft} - \bar{\theta}_{it})(\omega_{ft} - \bar{\omega}_{it}), \quad (5)$$

where  $\omega_{ft}$  is firm-level productivity,  $\theta_{ft}$  is the share of employment of the firm, and a “bar” over a variable represents the unweighted industry average of the firm-level measure. The covariance reflects the extent to which firms with higher than average productivity have a higher than average share of employment. Bartelsman et al. (2013) argue that it can be viewed as a measure of allocative efficiency. The rationale is that more resources should be allocated to more productive firms, so more productive firms should be larger. In an accounting sense, the covariance measures the difference between the actual industry productivity index and its counterfactual if employment shares were randomly allocated within industries. As firm productivity is measured in logs, a one percent increase of the covariance is associated with a one percent increase of industry productivity. Changes in the covariance thus measure changes in the industry productivity index due to changing allocative efficiency while holding firm productivity constant. We compute the covariance using the balance sheet information in our data. Firm productivity is estimated using the Olley and Pakes (1996) method.

## 5. Empirical evidence

This section presents the empirical evidence. We start with an overview of the data in sub-section 5.1, which is followed by formal econometric tests in sub-section 5.2. Sub-section 5.3 examines the parallel pre-trends of the GPL, and presents a series of robustness checks. Finally, we test other model predictions to provide empirical support to our model in sub-section 5.4.

<sup>27</sup> Table A.1 in the Appendix provides a detailed list of each two-digit industry's share of sales to government and its treatment status.

**Table 2**  
Summary Statistics.

| Variable                        | Obs  | Mean  | Std. Dev. | Min    | Max    |
|---------------------------------|------|-------|-----------|--------|--------|
| <i>Allocative efficiency</i>    | 4295 | -.213 | .315      | -2.579 | 1.416  |
| <i>gs</i>                       | 4295 | .671  | .47       | 0      | 1      |
| <i>post</i>                     | 4295 | .52   | .5        | 0      | 1      |
| <i>SOE share</i>                | 4295 | .143  | .159      | 0      | .977   |
| <i>FIE share</i>                | 4295 | .213  | .142      | 0      | .825   |
| <i>log import tariff</i>        | 4180 | 2.409 | .69       | -.693  | 4.174  |
| <i>interest rate dispersion</i> | 4295 | .345  | .914      | .016   | 7.014  |
| <i>tax rate dispersion</i>      | 4295 | .11   | .209      | .029   | 1.651  |
| <i>subsidy dispersion</i>       | 4295 | .036  | .07       | 0      | .522   |
| <i>log capital-labor ratio</i>  | 4295 | 4.278 | .636      | 2.205  | 6.936  |
| <i>HHI</i>                      | 4295 | .045  | .06       | .001   | .702   |
| <i>log number of firms</i>      | 4295 | 5.435 | 1.162     | 2.303  | 9.534  |
| <i>log sales</i>                | 4295 | 16.23 | 1.475     | 11.835 | 21.543 |

Note: *Allocation efficiency* is the size-productivity correlation measure; *gs* is a dummy variable for the treatment status; *post* is a dummy for years since 2003, year of the enactment of the GPL; *SOE share* and *FIE share* are share of the number of State Owned Enterprises and Foreign Invested Enterprises in each industry, respectively; *log import tariff* is the log of average import tariff; *interest rate dispersion*, *tax rate dispersion*, *subsidy dispersion* are the standard deviation of effective interest rate, tax rate, and subsidy intensity across firms within each industry; *log capital-labor ratio* is the log capital-labor ratio of the industry; *HHI* is the Herfindahl–Hirschman Index of sales of each industry; *log number of firms* and *log sales* are the log of total number of firms and total sales for each industry.

### 5.1. Summary statistics

Summary statistics of the main variables used in this analysis is presented in Table 2. Allocative efficiency as measured by the size–productivity covariance has sufficient variation across industries. The standard deviation of the covariance is 0.315, compared to the mean of -0.213.<sup>28</sup> Taking the simple average of the covariance across years, we find that the industry with the lowest allocative efficiency is the crude oil processing and petroleum products industry, which mainly consists of subsidiary firms of Sinopec and Petro China. The industry with the highest allocative efficiency is the home air conditioner industry, which is a very specialized industry consisting mostly of private and foreign-invested firms. The mean of the dummy for treatment status  $gs_i$  is 0.671, indicating that the treatment and control groups are roughly balanced, with 67.1% of all manufacturing industries in the treatment group. We also have a roughly half-half division of observations before and after the enactment of the GPL, with the mean of time dummy  $post_i$  at 0.52. Summary statistics of other control variables are also present. We will discuss these variables later when we present the relevant empirical results. Table A.2 in the appendix presents the corresponding correlation table for these variables.

### 5.2. Baseline results

We next examine the effect of the GPL formally. Table 3 reports our baseline empirical results. Column (1) shows the estimation results when only industry- and year-fixed effects are controlled for. The estimate of  $\beta$  is significant and positive, indicating that after the implementation of the GPL, the allocative efficiency of industries in the treatment group improves relative to that of industries in the control group. But this result may be biased due to omitted variables that affect resource allocation and correlate with our key explanatory variable at the same time. In the follow-up regressions, we control for these potential confounding factors step by step.

#### Simultaneous Policy Changes

The effect of the GPL can be confounded by other important policy changes that also took place in China around 2003. Important candidates include the SOE reform, the deregulation of foreign invested enterprises (FIE), and the accession to WTO in 2001. These policy changes have profound impacts on the Chinese economy, and may improve resource allocation as well. However, as long as the impact of these policies on different industries is not systematically related to the share of government procurement, they do not invalidate our identification. Nonetheless, we further control for these potential confounding factors to confirm that our results are not driven by them.

First, the SOE reform and the deregulation of FIE started before the enactment of GPL and accelerated afterward. The SOE reform resulted in large-scale privatization, the close-down of small SOEs, and improvement in the efficiency of surviving (large) SOEs. Relaxed entry regulation of foreign firms fostered the rise of FDI. These policy changes could as well improve resource allocation. For example, the close-down of low productivity small SOEs led to the reallocation of resources to productive private firms (Hsieh and Song, 2015). We capture their potential influences by including the share of SOEs and that of FIEs in each industry following Lu and Yu (2015) and Liu et al. (2019). Column (2) shows that our key estimate is robust to controlling for the SOE share. The SOE share is positively correlated with allocative efficiency. The SOE reform closed or privatized unproductive small SOEs but retained

<sup>28</sup> The covariance could be negative, indicating severe resource misallocation. Bartelsman et al. (2013) also find a negative covariance for Romania.

**Table 3**  
Baseline Results.

|                                 | (1)                  | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   | (7)                   | (8)                   | (9)                    | (10)                   | (11)                   |
|---------------------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|------------------------|------------------------|
| <i>gs*post</i>                  | 0.1310**<br>(0.0519) | 0.1376**<br>(0.0504)  | 0.1364**<br>(0.0509)  | 0.1376**<br>(0.0508)  | 0.1376**<br>(0.0507)  | 0.1373**<br>(0.0509)  | 0.1369**<br>(0.0507)  | 0.1369**<br>(0.0511)  | 0.1247**<br>(0.0472)   | 0.1241**<br>(0.0478)   | 0.1249**<br>(0.0485)   |
| <i>SOE share</i>                |                      | 0.4364***<br>(0.1282) | 0.4563***<br>(0.1330) | 0.4831***<br>(0.1303) | 0.4836***<br>(0.1300) | 0.4819***<br>(0.1301) | 0.4738***<br>(0.1289) | 0.4882***<br>(0.1375) | 0.4868***<br>(0.1344)  | 0.4142***<br>(0.1329)  | 0.4344***<br>(0.1413)  |
| <i>FIE share</i>                |                      |                       | 0.0862<br>(0.1442)    | 0.1458<br>(0.1309)    | 0.1460<br>(0.1308)    | 0.1443<br>(0.1306)    | 0.1426<br>(0.1302)    | 0.1437<br>(0.1307)    | 0.1604<br>(0.1288)     | 0.1416<br>(0.1258)     | 0.1159<br>(0.1254)     |
| <i>log import tariff</i>        |                      |                       |                       | -0.0621**<br>(0.0244) | -0.0623**<br>(0.0244) | -0.0619**<br>(0.0244) | -0.0616**<br>(0.0243) | -0.0625**<br>(0.0235) | -0.0616***<br>(0.0204) | -0.0598***<br>(0.0209) | -0.0605***<br>(0.0209) |
| <i>interest rate dispersion</i> |                      |                       |                       |                       | -0.0019<br>(0.0037)   | -0.0019<br>(0.0037)   | -0.0021<br>(0.0037)   | -0.0021<br>(0.0037)   | -0.0018<br>(0.0039)    | -0.0033<br>(0.0042)    | -0.0034<br>(0.0042)    |
| <i>tax rate dispersion</i>      |                      |                       |                       |                       |                       | 0.0157<br>(0.0124)    | 0.0128<br>(0.0125)    | 0.0123<br>(0.0123)    | 0.0238<br>(0.0142)     | 0.0174<br>(0.0132)     | 0.0179<br>(0.0134)     |
| <i>subsidy dispersion</i>       |                      |                       |                       |                       |                       |                       | 0.0818<br>(0.0561)    | 0.0807<br>(0.0566)    | 0.0750<br>(0.0518)     | 0.0585<br>(0.0552)     | 0.0599<br>(0.0559)     |
| <i>log capital-labor ratio</i>  |                      |                       |                       |                       |                       |                       |                       | 0.0218<br>(0.0353)    | -0.0138<br>(0.0363)    | -0.0147<br>(0.0378)    | -0.0300<br>(0.0346)    |
| <i>HHI</i>                      |                      |                       |                       |                       |                       |                       |                       |                       | 1.6053***<br>(0.3588)  | 1.8442***<br>(0.3602)  | 1.7162***<br>(0.4061)  |
| <i>log number of firms</i>      |                      |                       |                       |                       |                       |                       |                       |                       |                        | 0.1049***<br>(0.0278)  | 0.0576<br>(0.0572)     |
| <i>log sales</i>                |                      |                       |                       |                       |                       |                       |                       |                       |                        |                        | 0.0457<br>(0.0606)     |
| FE                              | Industry, Year       |                       |                       |                       |                       |                       |                       |                       |                        |                        |                        |
| N                               | 4295                 | 4295                  | 4295                  | 4180                  | 4180                  | 4180                  | 4180                  | 4180                  | 4180                   | 4180                   | 4180                   |

Note: Dependent variable is the size-productivity correlation measure of the allocative efficiency. Robust standard errors in parentheses, clustered at CIC four-digit industry level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

productive large firms. The remaining SOEs thus tend to be large and productive, which can explain the positive correlation. Column (3) further includes the share of FIEs into the regression, which does not significantly alter our estimates either. Our finding is thus not driven by the SOE reform and the deregulation of FIE.

Second, China joined the WTO in 2001 and drastically reduced the import tariffs in various industries in a short period of time. Lu and Yu (2015) show that the tariff reduction raises allocative efficiency: industries that experienced larger tariff reductions tend to have smaller markup dispersion after the WTO accession. To control for the effect of WTO accession, we include the average import tariff at the industry level for each year in the regression.<sup>29</sup> The empirical results in column (4) show that tariff reduction does not drive the effect of the GPL.

### Industrial Policies

In this paper, we try to show that the government can affect resource allocation by directly participating in the market as a buyer. Yet, the literature mainly emphasizes the government's role in allocating resources as a policy maker. To substantiate the government's impact on resource allocation as a buyer, we need to tease out the effect of various government policies.

Industrial policy is an important channel for the government to influence the market. Industrial policies are actively adopted by governments all over the world, and China is no exception. Such policies can affect within industry allocation of resources as they may not apply to all firms in an industry uniformly. For example, industrial policies may change the degree of competition in a market, which leads to changes in the markup dispersion (Aghion et al., 2015). Changing industrial policies during our sample period thus may contaminate our estimation results. Industrial policies are diverse and hard to measure. Aghion et al. (2015) argue that such policies often manifest as protective tariffs, low-interest loans, tax incentives, and subsidies in China. Following them, we control for these policy tools in an attempt to capture the effect of various industrial policies. We have shown that the tariffs do not affect our results in the previous regression; now we try to further control for the rest.

Low-interests loans provided by state-owned banks or government funds are used by the Chinese government to promote investments in key sectors. If some firms can obtain loans at prices lower than other firms, such policies can distort the allocation of capital between firms. We cannot directly observe loan prices faced by individual firms. Instead, we divide total interest expenditure by current liabilities to derive a measure of the effective interest rate of corporate loans. We include in our regression each industry's standard deviation of the effective interest rate to control for the potential distortionary effect of low-interest loans. A higher standard deviation indicates that some firms have much higher (lower) financing costs than others, suggesting potentially larger distortions in capital allocation. The results in column (5) show that the distribution of effective interest rate has no significant effect on allocative efficiency. The effect of the GPL still holds.

Differential tax treatment, through tax holidays or other tax subsidy programs, is another important policy tool of the government. Such policies lead to differences in effective tax rates across firms, which can be a source of misallocation (Bachas et al., 2019). Therefore, we add the within-industry standard deviation of firms' effective tax rate measured by the ratio of total tax payment to total sales, to control for the impact of tax policies. Column (6) shows that the effect of the GPL remains positive and significant after controlling for the variation in tax rates.

<sup>29</sup> As discussed later, tariffs can also be viewed as a type of industrial policy.

Similarly, subsidy is another commonly used tool to implement industrial policies. In column (7), we further control for the within-industry standard deviation of firms' subsidy intensity, measured by total subsidy over total sales. The positive effect of the GPL still holds.

In summary, although changing industrial policies could lead to changes in allocative efficiency over time, the policymaker role of the government does not diminish the market player's role.

### Industry Characteristics

Industries may differ in allocative efficiency because they are different in technology, industrial organization, and so on. Although we have controlled for industry-fixed effects in our previous regressions, industry characteristics may also be time-varying and confound our analysis. We next try to control for such confounding factors.

We first control for the capital–labor ratio of each industry. As the Chinese financial markets are under-developed, Chinese firms are often financially constrained (Song et al., 2011). Financial constraints are more distortionary in capital-intensive industries due to the higher need for capital investment (Buera et al., 2011). If industries relying more on government procurement also experienced a larger/smaller increase in capital intensity during our sample period,<sup>30</sup> our estimates of the effect of the GPL may be biased. Column (8) reports the results controlling for the industry capital–labor ratio. Our finding is robust to this concern.

We next add a set of controls that capture differences in industrial organization between industries. These include the Herfindahl–Hirschman Index (HHI) of the industry for each year (column (9)), the logarithm of total number of firms (column (10)), and the market size measured by the industry's total sales for each year (column (11)). With these controls, we hope to capture the degree of competition in different industries. Competition can asymmetrically change individual firm's pricing strategy and trigger entry–exit dynamics that alter the competing firms in an industry, both of which can lead to changes in allocative efficiency. The results in columns (9)–(11) show that the effect of GPL remains unchanged with these additional controls.

We thus confirm that our baseline result is not driven by obvious confounding factors. Lastly, we note that the effect of the GPL is economically significant. Take the estimate with the full set of controls (column (11)) as our estimate of the effect of the GPL. Compared with industries in the control group, the allocative efficiency of treated industries on average increases by 12.49% after the enactment of the GPL. As discussed above, this effect roughly translates into a 12.49% increase of the industry productivity index, which is remarkable even when compared with the fast Chinese economic growth during the period.

Our reduced-form analysis seems to have found a much larger effect of the GPL than that suggested by our numerical analysis at the end of Section 3. If we run a similar DID regression on simulated data from the numerical analysis, the effect of the GPL is 0.029 when we increase  $\theta$  to 28, slightly less than 1/4 of the reduced-form estimate. This does not indicate an issue with either of the analyses, because our simple calibration of the model is not intended to capture all the complexities in data and explain the observed changes in allocative efficiency in full. A major feature in the data that our simple calibration ignores is the presence of other distortions (Hsieh and Klenow, 2009). This probably explains why the size-productivity covariance implied by the model is much larger than that in the data. If different kinds of distortions reinforce each other, a reduction in the distortion of allocation of procurement orders could well induce larger changes in allocative efficiency. For example, firms mainly relying on receiving government orders might find it optimal to quit the market. If these firms also enjoy preferable treatments in financing, the distortionary effect of financial frictions also lowers after the GPL.

To examine the impact of other distortions, we extend our model to include a labor market friction. We assume the labor market friction follows a log-normal distribution and independent of other firm characteristics, which allows us to calibrate the variance of the friction to observed variation in firm labor productivity in untreated industries. With this simple extension, we find that the simulated effect of the GPL rises to 0.038, now close to 1/3 of the reduced-form estimate. Though it still falls short to explain the full effect, adding more realistic features to the model does improve the explanatory power of the model.

On the other hand, the numerical analysis does support us in finding significant results even if the Chinese reform might not be perfect. Starting a low value, even a small rise in  $\theta$  could raise the size-productivity covariance significantly. This is indeed what we find in the data, even though we can not claim allocation of procurement orders is free from distortions after the GPL. The fact that the Chinese government continues its reform of government procurement even until this day suggests that it is not the case.<sup>31</sup>

### 5.3. Validity and robustness

#### Parallel Pre-trends

To identify the causal effect of government procurement on resource allocation, our DID estimation compares the changes in allocative efficiency before and after the GPL between industries relying more on government procurement (treatment group) and industries relying less on procurement (control group). The validity of the empirical strategy relies on the assumption of parallel trends, that is, without the GPL, the control group and the treatment group should have the same trend in allocative efficiency.

<sup>30</sup> Different changes in capital intensity can be driven by differences in the elasticity of substitution between capital and labor, as shown in Alvarez-Cuadrado et al. (2017). They can also be induced by changes in industrial policies. If we fail to fully control for the effects of industrial policies above, further controlling for capital intensity helps resolve this problem.

<sup>31</sup> For example, the GPL was revised in 2014. The “Implementing Regulations of the Government Procurement Law of the People's Republic of China” was released in 2015. China continued to apply for accession to the Government Procurement Agreement, the most recent revised offer was submitted in October of 2019. The accession negotiations are still on-going.

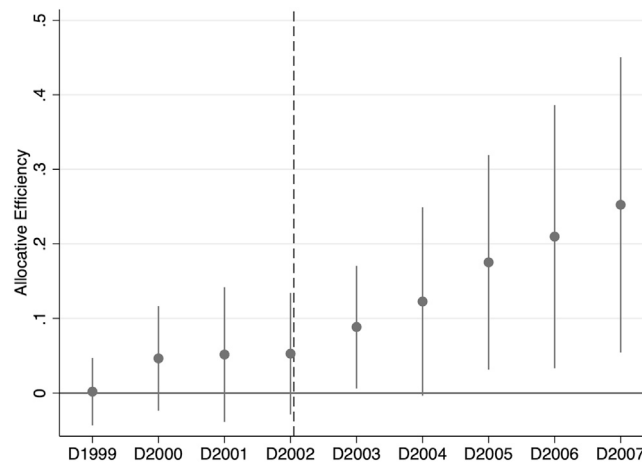


Fig. 3. Parallel Pre-trends.

To provide a rigorous test of parallel trends, we replace the post-reform dummy in the estimation equation with a vector of dummies for the years 1999–2007, so that we can see how the gap in allocative efficiency between the treatment group and the control group evolves relative to the base year (1998). We conduct the following flexible estimation:

$$y_{it} = \sum_i \beta_i \cdot gs_i \times year_t + \mathbf{X}_{it}' \gamma + \lambda_i + \lambda_t + \epsilon_{it}.$$

The time trend of the gap in allocative efficiency and the corresponding 95% confidence intervals are plotted in Fig. 3. The gap is insignificant and close to zero in 1999 and stays insignificant and stable during 2000–2002. It becomes significantly positive after 2003 and exhibits an increasing trend over time.<sup>32</sup> The results support that industries have parallel pre-trends, and resource allocation in industries relying more on government procurement improves after the GPL.

#### Measurement Errors in Government Procurement

Ideally, the treatment status should fully reflect an industry's consistent reliance on government procurement. However, potential measurement problems exist with our treatment status, which relies on firms' reported value of government procurement in 2004. We next discuss these problems in detail and provide some remedies.

Our treatment status can be mismeasured for several reasons. First, there may be measurement errors in the reported values of government procurement share by surveyed firms. These errors can add up to the industry level. Second, we use one year's data to construct government procurement share. This share may not correctly reflect an industry's reliance on government procurement if it is volatile over time. Third, we use 2004's data, which is after the GPL. One may have the concern that procurement shares in 2004 may have been altered by the GPL. Fourth, the government procurement shares are computed for two-digit industries and then matched to four-digit industries, for which we compute allocative efficiency. Our analysis allows for technology heterogeneity between four-digit industries within each two-digit industry in computing firm productivity.<sup>33</sup> Nevertheless, it may lead to mismeasured treatment status if there are substantial differences in government procurement shares between four-digit industries within a two-digit industry.

As the GPL mainly aims at standardizing and regulating the procurement procedures, it is reasonable to believe that the GPL mainly affects procurement efficiency within each industry rather than reallocates procurement orders across industries. The distribution of procurement across industries should mainly reflect demand factors driven by the public services the governments seek to provide. We thus believe the second and the third concerns may not be that important. Nevertheless, we have tried to reduce the impact of the first three measurement problems by classifying industries into a treated and a control group, instead of using a continuous treatment. The idea is that measurement errors in the procurement shares are less likely to change the ranking of industries too much. Fortunately, as discussed above, we find that the dichotomy of industries is consistent with our prior about what the government purchases and provides us with some confidence that these measurement errors should not pose a severe issue. To further relieve concerns about the mismeasured treatment status, we further present the following evidence.

First, we further compare the top 1/3 with bottom 1/3 industries in government procurement shares. Measurement errors should be even less likely to place an industry from the top (bottom) 1/3 to the bottom (top) 1/3. Defining  $gs3_i = 1$  for the top 1/3 industries and  $gs3_i = 0$  for the bottom 1/3 industries, we rerun our baseline specification by replacing  $gs_i$  with  $gs3_i$ . Column (1) of Table 4 presents the results. Evidently, the estimate of our interest stays significantly positive.

<sup>32</sup> The estimated coefficients are reported in Table A.3 in the Appendix.

<sup>33</sup> Since we assume Cobb–Douglas production functions in estimating firm productivity, technology heterogeneity in this paper means differences in the output elasticity of capital/labor.

**Table 4**  
Robustness: Different Measures of Treatment Status and Allocative Efficiency.

|                   | (1) 33%-33%         | (2) $gs_{io}$        | (3) 2-digit ind      | (4) TFPLP           | (5) 75-25th           |
|-------------------|---------------------|----------------------|----------------------|---------------------|-----------------------|
| $gs3^{*post}$     | 0.129**<br>(0.0611) |                      |                      |                     |                       |
| $gs_{io}^{*post}$ |                     | 0.172***<br>(0.0451) |                      |                     |                       |
| $gs^{*post}$      |                     |                      | 0.249***<br>(0.0757) | 0.0477*<br>(0.0265) | -0.0947**<br>(0.0427) |
| Controls          | Yes                 |                      |                      |                     |                       |
| FE                | Industry, Year      |                      |                      |                     |                       |
| N                 | 2884                | 4180                 | 290                  | 3827                | 4180                  |

Note:  $gs3$  is an alternative dummy variable for treatment status, with  $gs3=1$  if the industry's government procurement share is in the top 33%, and  $gs3=0$  if the share is at the bottom 33%;  $gs_{io}$  is an alternative dummy variable for treatment status based on government procurement share revealed in the Input Output (IO) Table, with  $gs_{io}=1$  if the industry's IO-based government procurement share is above the median, and  $gs_{io}=0$  otherwise. Controls include all control variables as in column (11) of Table 2. Columns (1)–(2) deal with potential measurement errors in the government procurement shares, (3)–(5) check robustness of alternative measures for allocative efficiency. Robust standard errors in parentheses, clustered at CIC four-digit industry level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Second, we use the 2003 CICS data to compute procurement shares in an earlier year. The 2003 survey reports government procurement shares in 2002, the year before the enactment of the GPL. However, it contains much fewer observations and only classifies firms into one-digit industries, including only six manufacturing sectors. We compute the procurement shares in 2004 for each one-digit industry and compare them with those in 2002. The correlation coefficient between the shares in 2002 and in 2004 is as high as 0.94, suggesting that the industry distribution of government procurement shares is stable over time.

Third, we propose an alternative measure of treatment status using the 2002 IO Table of China. The IO Table provides data aggregated at the industry level but does not directly contain information on government procurement. We use the purchase by the Public Administration and Social Organization (PASO) sector to proxy for government procurement. If the PASO sector purchases a positive amount from an industry, we treat the industry as a government procurement industry and define  $gs_{io_i} = 1$ ; otherwise, the industry is classified into the control group with  $gs_{io_i} = 0$ .<sup>34</sup> There are two advantages of this alternative measure. First, the IO data are formally estimated using comprehensive transactions data across industries by the National Bureau of Statistics of China, which should suffer less from reporting errors. Second, the purchase information was collected before 2002, and thus should not be altered by the GPL. A shortcoming of this measure is that purchases made by the PASO sector are not all subject to the GPL. Measurement errors in  $gs_{io_i}$  however are unlikely to be systematically correlated with measurement errors in  $gs_i$ . Table A.4 in the appendix tabulates  $gs_i$  and  $gs_{io_i}$ . Clearly, they largely overlap with each other, suggesting that both measures capture each industry's reliance on government procurement well. Column (2) of Table 4 presents results using  $gs_{io_i}$  instead of  $gs_i$ . The key estimate is still significantly positive.

Finally, we rerun the baseline regression at two-digit industry level to deal with the fourth concern. Doing so ensures that we measure the treatment status correctly but introduces errors in allocative efficiency because we now have to compute firm technology under the assumption that all firms within a two-digit industry operate with the same technology. The results in Column (3) show that the effect of the GPL remains significant and positive.

In summary, the additional evidence suggests that our finding is not likely to be driven by errors in treatment status.

#### Alternative Measures of Allocative Efficiency

In our baseline regressions, firm productivity is estimated using the Olley and Pakes (1996) method. We also re-examine our results using the Levinsohn and Petrin (2003) method to estimate productivity. The results in column (4) of Table 4 show that our finding is robust to this alternative measure.

We use the size–productivity covariance proposed by Bartelsman et al. (2013) to measure allocative efficiency, as suggested by our theoretical model. Another commonly used measure for misallocation is the dispersion of productivity (Hsieh and Klenow, 2009). We also check the robustness of our finding to this measure. Specifically, we compute the 75–25 percentile difference of firm productivity and use it as the dependent variable to rerun the baseline specification. The results in column (5) show that allocative efficiency measured by the productivity dispersion also improves in the treatment group after the GPL compared with the control group.

#### Expectation Effect

The GPL was approved in June 2002 and came into effect on January 1, 2003. In 2002, governments at all levels had clearly expected that the law would be implemented in 2003. Therefore, it is possible for them to change government procurement practice in the second half of 2002. Governments could standardize the procurement process in response to the law in advance, or they could speed up purchases that are difficult to carry out after the implementation of the law, or they could reduce procurement anticipating

<sup>34</sup> Excluding industries that only supply a negligible share of their output to the government (lower than 0.1% or 0.5%) from the treatment group does not affect the results.

**Table 5**  
Robustness: Further Concerns.

|                           | (1) Expectation      | (2) SOE Procure      | (3) Dom Firms        | (4) Pharmaceutical   | (5) Two-period      |
|---------------------------|----------------------|----------------------|----------------------|----------------------|---------------------|
| $gs_{i,t}^{post}$         | 0.1307**<br>(0.0495) | 0.1342**<br>(0.0532) | 0.1276**<br>(0.0507) | 0.1301**<br>(0.0524) | 0.125**<br>(0.0482) |
| $gs_{i,t}^{*}(year=2002)$ | 0.0292<br>(0.0229)   |                      |                      |                      |                     |
| Controls                  | Yes                  |                      |                      |                      |                     |
| FE                        | Industry, Year       |                      |                      |                      |                     |
| N                         | 4180                 | 4180                 | 4180                 | 4115                 | 806                 |

Note: Controls include all control variables as in column (11) of Table 2. Column (1) includes the expectation effect. Column (2) includes the procurement from SOEs as a control. Column (3) uses the subsample of domestic firms only. Column (4) excludes the pharmaceutical industry. Column (5) collapses the data to a two-period panel. Robust standard errors in parentheses, clustered at CIC four-digit industry level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

ongoing procurement might be subject to tighter regulation after 2003. In any case, we need to control for possible changes in government procurement practice in 2002. Therefore, we add  $gs_i \times 2002_t$  into our baseline specification to control for the potential expectation effect.

Column (1) of Table 5 presents the empirical results. The coefficient for  $gs_i \times 2002_t$  is positive but insignificant, showing that the expectation of the GPL, if any, had no significant effect before the law was formally enacted. The impact of the GPL remains positive and significant. Our finding is robust to the expectation effect.

#### SOE Procurement

As discussed in Section 2, the GPL only applies to procurement made by the government but not that made by SOEs, while SOE procurement can also be regarded as part of government procurement in the literature (USCBC, 2012). Instead, SOE procurement is subject to the Tendering and Bidding Law, which was enacted on January 1, 2000. Since our measure of government procurement share does not include SOE procurement, our estimate of the impact of the GPL thus should not be affected by the reform on SOE procurement. This can be seen from the parallel pre-trends test that there is no significant difference between the control group and the treatment group around 2000. Nevertheless, if the pattern of government procurement is similar to that of SOE procurement, SOE procurement still could be an important confounding factor. To eliminate this potential confounding effect, we further control for SOE procurement in our estimation. The CICS data also provide data on the percentage of sales to SOEs for each firm. We aggregate this information to reach an SOE procurement share for each industry, similar to the construction of the government procurement share. Analogously, we define a dummy variable according to whether the SOE procurement share in an industry is above the median.<sup>35</sup> We then add to our estimation the interaction of this dummy with a vector of yearly dummies, to fully control for the changing effect of SOE procurement over years. To save space, in the results presented in column (2) of Table 5, we do not report the coefficients of these interactions. We find that after controlling for the potential confounding effect from SOE procurement, the effect of the GPL is still significantly positive.

#### Sample of Domestic Firms

For various reasons, foreign firms often have restricted access to the government procurement market (USCBC, 2012). In the language of our model in Section 3, foreign firms have worse government connections than domestic firms. If the share of foreign firms increased over the years and the increase is systematically correlated with the treatment status,<sup>36</sup> our estimates can be biased. One source of the bias is that changes in the distribution of government connections due to the entry of foreign firms directly affect resource allocation in an industry. Besides that, allocative efficiency may change as foreign firms face a different business environment than domestic firms, such as better financing conditions and tighter regulations. Although we include the FIE share in our regression to partially alleviate this problem, we now further test the robustness of our results by using a sample of domestic firms. After excluding all FIEs from the sample, we recalculate allocative efficiency and all the corresponding controls for each industry and then re-estimate our benchmark regression. The results in column (3) of Table 5 confirm that our finding is robust and is not driven by FIEs.

#### Pharmaceutical Industry

The procurement of pharmaceutical products is subject to additional regulations, such as those issued by the National Healthcare Security Administration (USCBC, 2012). To tease out the potential confounding effect from this industry, we rerun our baseline regression using a sample excluding the pharmaceutical industry. The results in column (4) of Table 5 show that our finding is robust to this concern.

#### Serial Correlation

Serial correlation in the errors is a serious concern for DID inference as the resulting standard errors may be inconsistent (Bertrand et al., 2004). To alleviate this problem, in all regressions we have clustered the errors at the industry level, which allows for serial correlation and heteroscedasticity within each industry across years. To further confirm that serial correlation does not contaminate our finding, we follow Bertrand et al. (2004) to collapse the data into a pre- and a post-reform period and rerun our baseline

<sup>35</sup> Directly using shares instead of dummies gives similar results.

<sup>36</sup> For example, foreign firms in industries that rely on government procurement may be less willing to enter the Chinese market.

**Table 6**  
Mechanisms: Cross-section Evidence.

|                  | Firm Gov. Procurement Share |                       | Labor Productivity     |                        |                        |
|------------------|-----------------------------|-----------------------|------------------------|------------------------|------------------------|
|                  | (1)                         | (2)                   | (3)                    | (4)                    | (5)                    |
| <i>SOE</i>       | 0.0417***<br>(0.0060)       |                       |                        | -0.5842***<br>(0.0259) |                        |
| <i>ETC Share</i> |                             | 0.1507***<br>(0.0504) |                        |                        | -4.3120***<br>(0.4995) |
| <i>gs</i>        |                             |                       | -0.5060***<br>(0.0831) |                        |                        |
| Firm Sales       | No                          | No                    | Yes                    | Yes                    | Yes                    |
| FE               | Province*Industry           |                       |                        |                        |                        |
| N                | 12123                       | 12123                 | 12123                  | 12123                  | 12123                  |

Note: This table provides cross-section evidence for the mechanism using the firm-level CICS data. *SOE* is a dummy for firms being a state-owned enterprises; *ETC share* is the firm's share of entertainment and travel expenses over sales. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

regression. All variables are collapsed to their means for each period. The results in column (5) of Table 5 show that our estimate of interest remains positive and significant.

#### 5.4. Mechanism tests

We have shown that the enactment of the GPL increases allocative efficiency in industries relying more on government procurement, consistent with the main prediction of our model. This subsection further examines other empirical regularities predicted by the model using firm-level data. These tests provide empirical support for our model and allow us to corroborate the specific economic mechanism laid out in the model.

First, we examine whether there is a positive cross-section correlation between political connections ( $\eta$ ) and government procurement shares, as predicted in Proposition 1. We use SOE status as a proxy for government connections. SOEs are by nature better-connected to the government, as they are owned and managed by the state. In China, SOE executives also have government administrative ranks and often maintain close relationships with government officials. Many policies are also more inclined to SOEs. Therefore, it is reasonable to view SOEs as having a lower  $\eta$  than private firms. As the ASIFP does not have information on government procurement shares, we examine the correlation between SOE status and government procurement share in the CICS 2005 data. We include province-industry fixed effects to examine the within province-industry difference in procurement shares between SOEs and private firms. Column (1) of Table 6 shows that the effect of SOE status is significantly positive: procurement shares of SOEs on average are 4.0 percentage points higher than that of private firms. The positive correlation between political connections and government procurement shares is thus supported by data.

Second, Proposition 1 also predicts that better-connected firms devote more resources to acquiring government orders. We test for this prediction using enterprise entertainment and travel expenses (ETC) as a proxy for the resources devoted to acquiring government orders. Cai et al. (2011) argue that ETC reflects the degree of corruption of a firm because the expense includes the additional cost that the firm pays to obtain better government services, lower taxes, and so on. Procurement orders could well be another favor these firms seek. Also in the CICS 2005 data, we find that firms with higher ETC shares are more likely to obtain government orders in column (2). A one percentage point increase in the ETS share leads to a 1.1 percentage points increase in the government procurement share.

Our model further predicts that firms with higher government procurement shares tend to have lower productivity, conditional on firm sales (Proposition 2). The CICS 2005 data contains information on total sales, labor, wage bill but not capital. Therefore, we confine our test of this prediction to the correlation between procurement share and revenue labor productivity.<sup>37</sup> Specifically, we add up the wage payment to full-time and part-time labor to reach a measure of the total wage bill paid by a firm, and use the ratio of total sales to total wages as labor productivity ( $Y/L$ ).<sup>38</sup> The results in column (3) indicate that firms with a higher procurement share have significantly lower labor productivity. By Corollary 1, SOEs or firms with higher ETC shares should have lower labor productivity. This is indeed what we find in the CICS data (columns (4) and (5)).

Finally and more importantly, we provide evidence on the model mechanism using firm-level panel data. The mechanism behind the main prediction that the GPL raises the size–productivity covariance is that 1) firms with better government connections reduce their share of sales to the government, and 2) productivity in production for the government is also raised relative to that for the market. Both effects suggest that for firms with better government connections in industries that rely more on government orders, the enactment of the GPL would 1) reduce their sales, and 2) raise their productivity. Again using SOE status as a proxy of government connections, we can examine this prediction in the ASIFP data using the following triple-difference specification<sup>39</sup>:

<sup>37</sup> In the model labor is the only input such that firm TFP is the same as labor productivity. However, it should be understood that labor in the model represents all inputs such that in the empirical analysis, we mainly use firm TFP as our measure of productivity.

<sup>38</sup> We use wages to account for the differences in the quality of workers. Using the number of employees provides similar results.

<sup>39</sup> We restrict our sample to domestic firms as foreign-invested firms may have different trends in these outcomes.

**Table 7**  
Mechanisms: Panel Evidence.

|                                           | (1) lnSales                    | (2) TFP              | (3) lnSales                   | (4) TFP               | (5) lnSales          | (6) TFP              |
|-------------------------------------------|--------------------------------|----------------------|-------------------------------|-----------------------|----------------------|----------------------|
| $gs \cdot post \cdot SOE$                 | -0.0384***<br>(0.0126)         | 0.0380**<br>(0.0179) |                               |                       |                      |                      |
| $gs \cdot post \cdot lag \text{ subsidy}$ |                                |                      | -0.0177**<br>(0.0090)         | 0.0309***<br>(0.0114) | -0.0180*<br>(0.0101) | 0.0245**<br>(0.0120) |
| Firm Controls                             | Yes                            |                      |                               |                       |                      |                      |
| FE                                        | Firm, Ownership*Year, Ind*Year |                      | Firm, Province*Year, Ind*Year |                       |                      |                      |
| N                                         | 1566998                        | 1493415              | 1051963                       | 999340                | 945996               | 898929               |

Note: The sample includes manufacturing firms during 1998–2007 from ASIFP. *lag subsidy* is a dummy for whether the firm receives a subsidy in the previous year; Firm controls include firms' effective interest rate, effective tax rate, subsidy intensity, logarithm of capital-labor ratio, age, and an export dummy. Subsidy intensity is not included in the controls in columns (3)–(6). Robust standard errors in parentheses, clustered at firm level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

$$\omega_{ft} = \alpha \cdot gs_i \times post_t \times SOE_f + \mathbf{X}'_{ft} \delta + \lambda_f + \lambda_{ot} + \lambda_{it} + \epsilon_{fit},$$

where  $SOE_f$  is a dummy for SOE status.  $\omega_{ft}$  are firm-level outcomes predicted by the model, including the log of sales and productivity, as in the baseline regression. We control for important firm-level time-varying characteristics in  $\mathbf{X}_{ft}$ , including industrial policies such as the firms' effective interest rate, effective tax rate, and subsidy intensity. Mirroring the industry-level regression, we also include the logarithm of firms' capital-labor ratio in  $\mathbf{X}_{ft}$ . In addition, we add firm, ownership-year, and (two-digit) industry-year fixed effects to control for unobserved confounding factors. Robust standard errors are clustered at firm level. The estimate of  $\alpha$  measures the sales/productivity changes of SOEs relative to that of private firms after the GPL.

Columns (1) and (2) of Table 7 report results on firms' sales and productivity, respectively. The sales advantage of SOEs relative to non-SOEs decreases after the GPL in industries that rely on government procurement compared with that in industries that rely less on procurement, whereas the productivity of SOEs increases relative to non-SOEs in government procurement industries. The decrease in size and the increase in productivity of connected firms conform exactly with the economic mechanism revealed by our model, which rationalizes our baseline empirical findings.

Although SOEs and private firms differ in various ways, we are comparing the change in the difference between them before and after the reform across industries with varying dependence on government procurement. This level of variation specific to government procurement reliance, combined with our unique prediction of opposite reactions of revenue and productivity, has limited potential confounding factors. Nonetheless, we further redo our analysis using an alternative measure for political connection to ensure the robustness of our analysis. Firms tend to receive more subsidies when they are connected to the government (Jin and Zhang, 2019; Li et al., 2021), and the loss of such a connection will result in a substantial reduction in subsidies (Jin and Zhang, 2019). We thus define a dummy variable *lag subsidy*<sub>ft</sub> to index whether a firm received a subsidy in the previous period, and use it to proxy for the firm's government connection. We use the proxy to rerun the triple difference specification, controlling for province-year, industry-year, and firm fixed effects. The results are reported in Columns (3) and (4) in Table 7. Evidently, the results are consistent with our previous findings using SOE. To further tease out the possibility that the results are mainly driven by SOEs, we further run the regression excluding all SOEs. Results in Columns (5) and (6) show that our findings still hold.

## 6. Conclusion

This study examines the impact of the Chinese Government Procurement Law on resource misallocation within industries. We find that the GPL, which aims at curbing corruption in the government procurement market by standardizing procurement procedures, raises allocative efficiency in industries that rely more on government procurement, as measured by an increased covariance between firm size and productivity. Our findings are robust to a battery of robustness checks. We use a heterogenous firm model with government procurement to rationalize this finding. We also find empirical support for the model mechanisms.

Our study points to an important role of the government to interfere with resource allocation as a direct market participant, and suggests that inefficient government procurement may have deeper resource allocation implications than merely wasting public funds. Owing to data limitations, we are unable to examine the relationship between government connections, procurement orders, and firm productivity in a more detailed way. Future studies can improve on this front and further clarify the role of standardizing procurement procedures in improving allocative efficiency.

## Declaration of competing interest

Finally, this manuscript is original, no part of it has been published before, and there are no conflicts of interest to disclose.

## Data availability

The authors do not have permission to share data.

## Acknowledgement

We thank seminar participants at various institutions for their helpful suggestions and comments. We thank National Natural Science Foundation of China (No. 72003208 and 71903203), National Social Science Foundation of China (No. 22&ZD058 and 23ZDA052), and Guangdong Provincial Natural Science Foundation (No. 2022A1515012092) for financial support.

## Appendix A

**Table A.1**  
Classification of Industries.

| 2-digit Index | Description                                                       | Government Procurement Share |
|---------------|-------------------------------------------------------------------|------------------------------|
| 37            | Transport and communication facilities                            | 0.0749149                    |
| 18            | Textile costumes, shoes, and caps                                 | 0.0461449                    |
| 41            | Instruments and meters, and machinery for culture and office      | 0.035444                     |
| 31            | Non-metallic minerals product                                     | 0.0341722                    |
| 36            | Specialized facility                                              | 0.0332038                    |
| 42            | Artwork and other                                                 | 0.0322581                    |
| 26            | Chemical feedstock and chemical                                   | 0.0310103                    |
| 16            | Tobacco                                                           | 0.0299511                    |
| 15            | Beverage                                                          | 0.0225519                    |
| 20            | Wood processing and wood, bamboo, rattan, palm articles           | 0.0221124                    |
| 35            | General-purpose equipment                                         | 0.0200366                    |
| 39            | Electric machinery and equipment                                  | 0.0196648                    |
| 34            | Metalwork                                                         | 0.0187107                    |
| 32            | Ferrous metal smelting and extrusion                              | 0.0180803                    |
| 40            | Communication equipment, computers and other electronic equipment | 0.0180738                    |
| 21            | Cabinetmaking                                                     | 0.0171022                    |
| 27            | Medicine                                                          | 0.0156981                    |
| 23            | Printing and reproduction of record media                         | 0.0102881                    |
| 29            | Rubber                                                            | 0.0100021                    |
| 30            | Plastic                                                           | 0.0086541                    |
| 25            | Petroleum processing, coking and nuclear fuel manufacture         | 0.0082873                    |
| 22            | Papermaking and paper product                                     | 0.0082778                    |
| 33            | Non-ferrous smelting and extrusion                                | 0.0079693                    |
| 13            | Agricultural and non-staple foodstuff                             | 0.0078963                    |
| 17            | Textile                                                           | 0.0068004                    |
| 28            | Chemical fiber                                                    | 0.0062277                    |
| 19            | Leather, fur, feather (cloth with soft nap) and their products    | 0.0055399                    |
| 24            | Culture, education and sports goods                               | 0.0022037                    |
| 14            | Foodstuff                                                         | 0.0013263                    |
| 43            | Waste                                                             | 0                            |

**Table A.2**  
Correlations.

| Variables                    | (1)    | (2)    | (3)    | (4)    | (5)    | (6)    | (7)    | (8)    | (9)    | (10)  | (11)   | (12)  | (13)  |
|------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|-------|-------|
| (1) Allocative Efficiency    | 1.000  |        |        |        |        |        |        |        |        |       |        |       |       |
| (2) gs                       | 0.116  | 1.000  |        |        |        |        |        |        |        |       |        |       |       |
| (3) post                     | 0.107  | 0.008  | 1.000  |        |        |        |        |        |        |       |        |       |       |
| (4) SOE share                | -0.034 | 0.116  | -0.510 | 1.000  |        |        |        |        |        |       |        |       |       |
| (5) FIE share                | 0.149  | -0.126 | 0.066  | -0.425 | 1.000  |        |        |        |        |       |        |       |       |
| (6) log import tariff        | 0.142  | -0.107 | -0.326 | 0.090  | 0.162  | 1.000  |        |        |        |       |        |       |       |
| (7) interest rate dispersion | -0.014 | -0.053 | 0.111  | -0.080 | -0.013 | -0.015 | 1.000  |        |        |       |        |       |       |
| (8) tax rate dispersion      | 0.007  | 0.006  | -0.098 | 0.108  | -0.011 | 0.052  | 0.003  | 1.000  |        |       |        |       |       |
| (9) subsidy dispersion       | -0.059 | -0.006 | -0.164 | 0.239  | -0.108 | -0.008 | -0.002 | 0.146  | 1.000  |       |        |       |       |
| (10) log capital-labor ratio | -0.105 | -0.027 | 0.212  | -0.045 | -0.154 | -0.192 | 0.064  | 0.002  | 0.013  | 1.000 |        |       |       |
| (11) HHI                     | 0.145  | 0.049  | -0.074 | 0.171  | -0.013 | -0.021 | -0.093 | -0.036 | -0.008 | 0.086 | 1.000  |       |       |
| (12) log number of firms     | 0.014  | -0.079 | 0.184  | -0.303 | 0.016  | 0.014  | 0.181  | 0.082  | 0.020  | 0.063 | -0.595 | 1.000 |       |
| (13) log sales               | 0.059  | -0.040 | 0.306  | -0.316 | 0.089  | -0.073 | 0.163  | 0.051  | -0.003 | 0.408 | -0.355 | 0.817 | 1.000 |

## References

- Adamopoulos, T., Brandt, L., Leight, J., Restuccia, D., 2022. Misallocation, selection and productivity: a quantitative analysis with panel data from China. *Econometrica* 90 (3), 1261–1282.
- Adamopoulos, T., Restuccia, D., 2020. Land reform and productivity: a quantitative analysis with micro data. *Am. Econ. J. Macroecon.* 12 (3), 1–39.
- Aghion, P., Cai, J., Dewatripont, M., Du, L., Harrison, A., Legros, P., 2015. Industrial policy and competition. *Am. Econ. J. Macroecon.* 7 (4), 1–32.
- Alvarez-Cuadrado, F., Van Long, N., Poschke, M., 2017. Capital-labor substitution, structural change, and growth. *Theor. Econ.* 12 (3), 1229–1266.
- Andrews, D., Cingano, F., Conconi, P., 2014. Public policy and resource allocation: evidence from firms in OECD countries. *Econ. Policy* 29 (78), 255–296.
- Auriol, E., Straub, S., Flochel, T., 2016. Public procurement and rent-seeking: the case of Paraguay. *World Dev.* 77, 395–407.
- Bachas, P., Jaef, R.N.F., Jensen, A., 2019. Size-dependent tax enforcement and compliance: global evidence and aggregate implications. *J. Dev. Econ.* 140, 203–222.

**Table A.3**  
Flexible Estimation.

|                  | (1) Flexible         |
|------------------|----------------------|
| gs*(year = 1999) | 0.0034<br>(0.0213)   |
| gs*(year = 2000) | 0.0430<br>(0.0342)   |
| gs*(year = 2001) | 0.0496<br>(0.0438)   |
| gs*(year = 2002) | 0.0530<br>(0.0374)   |
| gs*(year = 2003) | 0.0759*<br>(0.0383)  |
| gs*(year = 2004) | 0.1157**<br>(0.0548) |
| gs*(year = 2005) | 0.1590**<br>(0.0622) |
| gs*(year = 2006) | 0.1948**<br>(0.0771) |
| gs*(year = 2007) | 0.2363**<br>(0.0859) |
| Controls         | Yes                  |
| FE               | Industry, Year       |
| N                | 4180                 |

Note: Controls include all control variables as in column (11) of Table 2. Robust standard errors in parentheses, clustered at CIC four-digit industry level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table A.4**  
Tabulation of  $gs$  and  $gs_{io}$ .

| gs_io | gs   |      |       |
|-------|------|------|-------|
|       | 0    | 1    | Total |
| 0     | 575  | 385  | 960   |
| 1     | 850  | 2495 | 3345  |
| Total | 1425 | 2880 | 4305  |

- Bai, C.-E., Hsieh, C.-T., Song, Z., 2020. Special deals with Chinese characteristics. *NBER Macroecon. Annu.* 34 (1), 341–379.
- Bai, C.-E., Li, D.D., Tao, Z., Wang, Y., 2000. A multitask theory of state enterprise reform. *J. Comp. Econ.* 28 (4), 716–738.
- Bai, C.-E., Lu, J., Tao, Z., 2006. The multitask theory of state enterprise reform: empirical evidence from China. *Am. Econ. Rev.* 96 (2), 353–357.
- Baltrunaite, A., 2019. Political contributions and public procurement: evidence from Lithuania. *J. Eur. Econ. Assoc.* 18 (2), 541–582.
- Bandiera, O., Prat, A., Valletti, T., 2009. Active and passive waste in government spending: evidence from a policy experiment. *Am. Econ. Rev.* 99 (4), 1278–1308.
- Baranek, B., Titl, V., 2020. The Cost of Favoritism in Public Procurement. *FEB Research Report Department of Economics*.
- Bartelsman, E., Haltiwanger, J., Scarpetta, S., 2013. Cross-country differences in productivity: the role of allocation and selection. *Am. Econ. Rev.* 103 (1), 305–334.
- Basu, S., 2019. Are price-cost markups rising in the United States? A discussion of the evidence. *J. Econ. Perspect.* 33 (3), 3–22.
- Baumol, W.J., 1990. Entrepreneurship: productive, unproductive, and destructive. *J. Polit. Econ.* 98 (5), 893–921.
- Bertrand, M., Duflo, E., Mullainathan, S., 2004. How much should we trust differences-in-differences estimates? *Q. J. Econ.* 119 (1), 249–275.
- Bosio, E., Djankov, S., Glaeser, E., Shleifer, A., 2022. Public procurement in law and in practice. *Am. Econ. Rev.* 112 (4), 1091–1117.
- Brandt, L., Tombe, T., Zhu, X., 2013. Factor market distortions across time, space and sectors in China. *Rev. Econ. Dyn.* 16 (1), 39–58.
- Brandt, L., Van Biesebroeck, J., Zhang, Y., 2012. Creative accounting or creative destruction? Firm-level productivity growth in Chinese manufacturing. *J. Dev. Econ.* 97 (2), 339–351.
- Broda, C., Weinstein, D.E., 2006. Globalization and the gains from variety. *Q. J. Econ.* 121 (2), 541–585.
- Brogaard, J., Denes, M., Duchin, R., 2020. Political influence and the renegotiation of government contracts. *Rev. Financ. Stud.* 34 (6), 3095–3137.
- Brugués, F., Brugués, J., Giambra, S., 2022. Political Connections and Misallocation of Procurement Contracts: Evidence from Ecuador. Working Paper.
- Buera, F.J., Kaboski, J.P., Shin, Y., 2011. Finance and development: a tale of two sectors. *Am. Econ. Rev.* 101 (5), 1964–2002.
- Cai, H., Fang, H., Xu, L.C., 2011. Eat, drink, firms, government: an investigation of corruption from the entertainment and travel costs of Chinese firms. *J. Law Econ.* 54 (1), 55–78.
- Cai, H., Liu, Q., 2009. Competition and corporate tax avoidance: evidence from Chinese industrial firms. *Econ. J.* 119 (537), 764–795.
- Cappelletti, M., Giuffrida, L.M., 2021. Procuring survival. *SSRN Electron. J.*
- Cheremukhin, A., Golosov, M., Guriev, S., Tsyvinski, A., 2015. The Economy of People's Republic of China from 1953. Working Paper.
- Cohen, L., Malloy, C.J., 2016. Mini West Virginias: corporations as government dependents. *SSRN Electron. J.*
- Coviello, D., Gagliarducci, S., 2017. Tenure in office and public procurement. *Am. Econ. J.: Econ. Policy* 9 (3), 59–105.
- Coviello, D., Marinello, M., 2014. Publicity requirements in public procurement: evidence from a regression discontinuity design. *J. Public Econ.* 109, 76–100.
- Czarnitzki, D., Hünermund, P., Moshgbar, N., 2018. Public Procurement as Policy Instrument for Innovation. Working Paper.
- di Giovanni, J., García-Santana, M., Jeena, P., Moral-Benito, E., Pijoan-Mas, J., 2022. Government procurement and access to credit: firm dynamics and aggregate implications. *SSRN Electron. J.*

- Ferraz, C., Finan, F., Szerman, D., 2015. Procuring Firm Growth: the Effects of Government Purchases on Firm Dynamics. Working Paper.
- Foster, L., Haltiwanger, J., Syverson, C., 2008. Reallocation, firm turnover, and efficiency: selection on productivity or profitability? *Am. Econ. Rev.* 98 (1), 394–425.
- Gabaix, X., 2016. Power laws in economics: an introduction. *J. Econ. Perspect.* 30 (1), 185–206.
- Garcia-Santana, M., Pijoan-Mas, J., 2014. The reservation laws in India and the misallocation of production factors. *J. Monet. Econ.* 66, 193–209.
- Garicano, L., Lelarge, C., Reenen, J.V., 2016. Firm size distortions and the productivity distribution: evidence from France. *Am. Econ. Rev.* 116 (11), 3439–3479.
- Goldman, E., Rocholl, J., So, J., 2013. Politically connected boards of directors and the allocation of procurement contracts. *Rev. Finance* 17 (5), 1617–1648.
- Gourio, F., Roys, N., 2014. Size-dependent regulations, firm size distribution, and reallocation. *Quant. Econ.* 5 (2), 377–416.
- Gugler, K., Weichselbaumer, M., Zulehner, C., 2020. Employment behavior and the economic crisis: evidence from winners and runners-up in procurement auctions. *J. Public Econ.* 182, 104112.
- Hebous, S., Zimmermann, T., 2021. Can government demand stimulate private investment? Evidence from U.S. federal procurement. *J. Monet. Econ.* 118, 178–194.
- Hopenhayn, H.A., 2014. Firms, misallocation, and aggregate productivity: a review. *Annu. Rev. Econ.* 6 (1), 735–770.
- Hsieh, C.-T., Klenow, P.J., 2009. Misallocation and manufacturing TFP in China and India. *Q. J. Econ.* 124 (4), 1403–1448.
- Hsieh, C.-T., Song, Z., 2015. Grasp the large, let go of the small: the transformation of the state sector in China. *Brookings Pap. Econ. Act.* 295.
- Jin, S., Zhang, Z., 2019. Political connections, government subsidies, and capital misallocation: evidence from China. Available at SSRN: <https://ssrn.com/abstract=3045703>.
- Lee, M., 2021. Government Purchases and Firm Growth. Working Paper.
- Levinsohn, J., Petrin, A., 2003. Estimating production functions using inputs to control for unobservables. *Rev. Econ. Stud.* 70 (2), 317–341.
- Li, X.-L., Li, J., Wang, J., Si, D.-K., 2021. Trade policy uncertainty, political connection and government subsidy: evidence from Chinese energy firms. *Energy Econ.* 99, 105272.
- Liu, Q., Qiu, L.D., Zhan, C., 2019. Trade liberalization and domestic vertical integration: evidence from China. *J. Int. Econ.* 121, 103250.
- Lu, Y., Yu, L., 2015. Trade liberalization and markup dispersion: evidence from China's WTO accession. *Am. Econ. J. Appl. Econ.* 7 (4), 221–253.
- Mironov, M., Zhuravskaya, E., 2016. Corruption in procurement and the political cycle in tunneling: evidence from financial transactions data. *Am. Econ. J.: Econ. Policy* 8 (2), 287–321.
- Mulabdic, A., Rotunno, L., 2021. Trade Barriers in Government Procurement. World Bank Policy Research Working Paper 9602.
- Neu, D., Everett, J., Rahaman, A.S., 2015. Preventing corruption within government procurement: constructing the disciplined and ethical subject. *Crit. Perspect. Account.* 28, 49–61.
- Olley, G.S., Pakes, A., 1996. The dynamics of productivity in the telecommunications equipment industry. *Econometrica* 64 (6), 1263–1297.
- Restuccia, D., Rogerson, R., 2008. Policy distortions and aggregate productivity with heterogeneous establishments. *Rev. Econ. Dyn.* 11, 707–720.
- Restuccia, D., Rogerson, R., 2017. The causes and costs of misallocation. *J. Econ. Perspect.* 31 (3), 151–174.
- Schoenherr, D., 2019. Political connections and allocative distortions. *J. Finance* 74 (2), 543–586.
- Slavtchev, V., Wiederhold, S., 2016. Does the technological content of government demand matter for private R&D? Evidence from US states. *Am. Econ. J. Macroecon.* 8 (2), 45–84.
- Song, Z., Storesletten, K., Zilibotti, F., 2011. Growing like China. *Am. Econ. Rev.* 101 (1), 196–233.
- Sraer, D., Thesmar, D., 2023. How to use natural experiments to estimate misallocation. *Am. Econ. Rev.* 113 (4), 906–938.
- Titl, V., Geys, B., 2019. Political donations and the allocation of public procurement contracts. *Eur. Econ. Rev.* 111, 443–458.
- Tombe, T., Winter, J., 2015. Environmental policy and misallocation: the productivity effect of intensity standards. *J. Environ. Econ. Manag.* 72, 137–163.
- Tombe, T., Zhu, X., 2019. Trade, migration, and productivity: a quantitative analysis of China. *Am. Econ. Rev.* 109 (5), 1843–1872.
- USCBC, 2012. Navigating China's Public Procurement Market: Background, Challenges, and Best Practices. The US-China Business Council.
- Young, A., 2000. The razor's edge: distortions and incremental reform in the People's Republic of China. *Q. J. Econ.* 115 (4), 1091–1135.